Integrating HIV, viral hepatitis and sexually transmitted infections with primary health care

Learning from countries



World Health Organization

Integrating HIV, viral hepatitis and sexually transmitted infections with primary health care

Learning from countries

World Health Organization

Integrating HIV, viral hepatitis and sexually transmitted infections with primary health care. Learning from countries

IISBN 978-92-4-011295-7 (electronic version)
ISBN 978-92-4-011296-4 (print version)

© World Health Organization 2025

Some rights reserved. This work is available under the Creative Commons Attribution-NonCommercial-ShareAlike 3.0 IGO licence (CC BY-NC-SA 3.0 IGO; https://creativecommons.org/licenses/by-nc-sa/3.0/igo).

Under the terms of this licence, you may copy, redistribute and adapt the work for non-commercial purposes, provided the work is appropriately cited, as indicated below. In any use of this work, there should be no suggestion that WHO endorses any specific organization, products or services. The use of the WHO logo is not permitted. If you adapt the work, then you must license your work under the same or equivalent Creative Commons licence. If you create a translation of this work, you should add the following disclaimer along with the suggested citation: “This translation was not created by the World Health Organization (WHO). WHO is not responsible for the content or accuracy of this translation. The original English edition shall be the binding and authentic edition”.

Any mediation relating to disputes arising under the licence shall be conducted in accordance with the mediation rules of the World Intellectual Property Organization (http://www.wipo.int/amc/en/mediation/rules/).

Suggested citation. Integrating HIV, viral hepatitis and sexually transmitted infections with primary health care. Learning from countries. Geneva: World Health Organization; 2025. Licence: CC BY-NC-SA 3.0 IGO.

Cataloguing-in-Publication (CIP) data. CIP data are available at https://iris.who.int/.

Sales, rights and licensing. To purchase WHO publications, see https://www.who.int/publications/book-orders. To submit requests for commercial use and queries on rights and licensing, see https://www.who.int/copyright.

Third-party materials. If you wish to reuse material from this work that is attributed to a third party, such as tables, figures or images, it is your responsibility to determine whether permission is needed for that reuse and to obtain permission from the copyright holder. The risk of claims resulting from infringement of any third-party-owned component in the work rests solely with the user.

General disclaimers. The designations employed and the presentation of the material in this publication do not imply the expression of any opinion whatsoever on the part of WHO concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries. Dotted and dashed lines on maps represent approximate border lines for which there may not yet be full agreement.

The mention of specific companies or of certain manufacturers’ products does not imply that they are endorsed or recommended by WHO in preference to others of a similar nature that are not mentioned. Errors and omissions excepted, the names of proprietary products are distinguished by initial capital letters.

All reasonable precautions have been taken by WHO to verify the information contained in this publication. However, the published material is being distributed without warranty of any kind, either expressed or implied. The responsibility for the interpretation and use of the material lies with the reader. In no event shall WHO be liable for damages arising from its use.

Design and layout: blossom.it

Contents

Acknowledgements iv

Glossary v

Background and scope 1

Section One 6

Angola 10
Botswana 11
Pakistan 12
Brazil 14
Kenya 15
Viet Nam 17

Section Two 19

Ethiopia 23
Indonesia 28
Rwanda 31
Zambia 34

Conclusions 38

Recommendations 39

References 40

Annex 42

Acknowledgements

This policy brief was developed by the World Health Organization (WHO) Department of Global HIV, Hepatitis and Sexually Transmitted Infections Programmes in consultation with the WHO Special Programme on Primary Health Care. Nirina Razakasoa, Andy Seale and Mukta Sharma of the Department of Global HIV, Hepatitis and Sexually Transmitted Infections Programmes led the development of the document with guidance from Shamsuzzoha Babar Syed of the WHO Special Programme on Primary Health Care. The following WHO staff provided input that informed the publication: Seblewongel Abate Nigussie, Rachel Baggaley, Stela Bivol, Budiarto, Lastone Chitembo, Carlos Cisneros, Meg Doherty, Kayita Godfrey, Heather Ingold, Tauhid Islam, Kiyohiko Izumi, Olufunmilayo Lesi, Antons Mozalevskis, Ribakare Muhayimpundu, Debora Pedrazzoli, Georges Perrin, Van Thi Thuy Nguyen, Peter Nsubuga, Susan Sparkes, Sarkar Swarup, Tova Tampe and Jiemei Chan.

The project also benefitted from significant support from Davina Canagasabey and Kimberly Green from PATH and Florence Riako Anam from the Global Network of People Living with HIV, organizations in official relations with WHO.

WHO also expresses appreciation to Ani Shakarishvili, UNAIDS, for significant input, expert review and guidance.

Finally, WHO thanks the partners engaged in the work to develop the country case studies described in Section Two of the policy brief and expresses appreciation for their contributions:

- Ethiopia's Ministry of Health, development partners and community organizations in driving the integration of health services;
- Indonesia's Ministry of Health, FHI 360, the WHO Country Office in Indonesia, the Global Fund to Fight AIDS, Tuberculosis and Malaria, the United States Agency for International Development and various community health workers and nongovernmental organizations that provided insight and access to documents;
- Rwanda's Ministry of Health, Rwanda Biomedical Centre, Rwanda Network of People Living with HIV, Rwanda Organization for Fighting Against Hepatitis, AIDS Health Foundation, Clinton Health Access Initiative, health-care workers and international partners and the communities involved in Rwanda's integrated health services; and
- Zambia's Ministry of Health, community health workers, international partners and the communities involved in the integration process in Zambia.

Appreciation is also extended to the following experts for final peer review: Serge Billong, Cordelia Katureebe and Bettina Schunter.

This brief was made possible through funding support from the UNAIDS Joint Programme, Unified Budget, Results and Accountability Framework (UBRAF) 2024 allocation to WHO.

Glossary[1]

Community health workers. Health workers who have received standardized and nationally endorsed training outside the nursing, midwifery or medical curricula.

Community-led AIDS responses. Actions and strategies that seek to improve the health and human rights of their constituencies, specifically informed and implemented by and for communities themselves and the organizations, groups and networks that represent them.

Community-led organizations, groups and networks. Whether formally or informally organized, these are entities for which most of the governance, leadership, staff, spokespeople, membership and volunteers reflect and represent the experiences, perspectives and voices of their constituencies and that have transparent mechanisms of accountability to their constituencies. Community-led organizations, groups and networks are self-determining and autonomous and not influenced by government, commercial or donor agendas. Not all community-based organizations are community-led.

Health system. All organizations, people and actions with the primary intent of promoting, restoring or maintaining health. This includes efforts to influence determinants of health and more direct health-improving activities. A health system is therefore more than the pyramid of publicly owned facilities that deliver personal health services. It includes, for example, family caregivers; private providers; behaviour change programmes; vector-control campaigns; health insurance organizations; and occupational health and safety legislation. The WHO health system framework identifies six health system building blocks: leadership and governance, health financing, health workforce, health services, health information systems and medical products, vaccines and technologies.

Integrated health services. Health services that are managed and delivered so that people receive a continuum of health promotion, disease prevention, diagnosis, treatment, disease management, rehabilitation and palliative care services coordinated across the different levels and sites of care within and beyond the health sector and according to their needs throughout the life-course.

Key populations. Groups that have a high risk and disproportionate burden of HIV in all epidemic settings. These groups frequently face legal and social challenges that increase their vulnerability to HIV, including barriers to accessing HIV prevention, diagnosis, treatment and other health and social services. Key populations include gay men and other men who have sex with men, people who inject drugs, people in prisons and closed settings, sex workers and transgender and gender-diverse people. Key populations often also face a disproportionate burden of viral hepatitis and sexually transmitted infections.

Lay provider. Any person who performs functions related to health-care delivery and has been trained to deliver specific services but has not received a formal professional or paraprofessional certificate or tertiary degree.

Person-centred care. Care approaches and practices in which the person is seen as a whole, with many levels of needs and goals, the needs being derived from their personal social determinants of health.

People-centred care. An approach to care that consciously adopts the perspectives of individuals, caregivers, families and communities as participants in and beneficiaries of trusted health systems organized around the comprehensive needs of people rather than individual diseases that respect social preferences. People-centred care is broader than patient- and person-centred care, encompassing not only clinical encounters but also including attention to the health of people in their communities and their crucial role in shaping health policy and health services.

1 Adapted from: Primary health care and HIV: convergent actions: policy considerations for decision-makers. Geneva: World Health Organization; 2023 (https://iris.who.int/handle/10665/371494, accessed 2 June 2025). Licence: CC BY-NC-SA 3.0 IGO.

Primary care. A key process in the health system that supports first-contact, accessible, continued, comprehensive and coordinated patient-focused care.

Primary health care. A whole-of-society approach to health that aims to maximize the level and distribution of health and well-being through three components: (1) primary care and essential public health functions as the core of integrated health services; (2) multisectoral policy and action; and (3) empowered people and communities.

Primary health care–oriented health system. A health system organized and operated to guarantee the right to the highest attainable level of health as the main goal while maximizing equity and solidarity. A primary health care–oriented health system comprises a core set of structural and functional elements that support achieving universal health coverage and access to services that are acceptable to the population and enhance equity.

Service package. A list of priority interventions and services across the continuum of care that should be made available to all individuals in a defined population. It may be endorsed by the government at the national or subnational levels or agreed by actors with services provided by a non-state actor.

Settings and sites of care. The varied types of arrangements for service delivery, organized further into facilities, institutions and organizations that provide care. Settings include ambulatory, community, home, inpatient and residential services, whereas facilities refer to infrastructure, such as clinics, health centres, district hospitals, dispensaries or other entities, such as mobile clinics and pharmacies.

Universal health coverage. Access for all people to the full range of high-quality health services they need, when and where they need them, without financial hardship. Universal health coverage encompasses the continuum of essential health services, from health promotion to disease prevention, treatment, rehabilitation and palliative care across the life-course.

Vulnerable populations. Groups of people who are vulnerable to HIV, hepatitis or sexually transmitted infections in certain situations or contexts, such as infants, children and adolescents (including adolescent girls in sub-Saharan Africa), orphans, people with disabilities and migrant and mobile workers. They may also face social and legal barriers to accessing HIV prevention and treatment. These populations are not affected by HIV uniformly in all countries, and epidemics and may include key populations, indigenous people and ethnic minorities.

Background and scope



Introduction

HIV, viral hepatitis and sexually transmitted infections (STIs) remain significant global health challenges, causing millions of deaths and new infections each year.

Despite important progress, the key targets and goals outlined in the integrated global health sector strategies on HIV, viral hepatitis and STIs for 2022–2030 *(1)* are unlikely to be reached.

There has been substantial progress in country adoption of universal health coverage goals and integrated, people-centred primary health care (PHC) as the approach to attain universal health coverage. Domestic reforms and financing for universal health coverage and PHC accelerated during and just after the coronavirus disease 2019 (COVID-19) emergency response. Although these reforms are under threat because of reduced international funding, they also offer a blueprint for sustainability and a potential platform for integrating HIV, STI and viral hepatitis services.

WHO and its partners have long identified the excessive reliance on a small number of donors as a threat to country-owned and sustainable programmes. Strategic frameworks, policy documents and technical normative guidance promoting integration and convergence with efforts to strengthen health systems and PHC, developed in recent years, have been used to support countries to adapt more sustainable and people-centred approaches. Indeed, WHO worked with the Joint United Nations Programme on HIV/AIDS (UNAIDS) and the Global Fund to Fight AIDS, Tuberculosis and Malaria to support countries in developing sustainability roadmaps for the HIV response throughout 2024 and early 2025 and to document and share best practices through webinars and conference sessions.

Although maintaining a disease-specific focus helps to drive progress towards impact, organizing services around the holistic needs of individuals and communities rather than around specific disease outcomes is also important. This has been challenging when numerous funding sources have maintained a disease-specific focus.

Promoting integrated person-centred approaches and convergence with PHC is at the heart of the global health sector strategies. The strategies promote the importance of creating the right balance between maintaining a disease-specific focus while simultaneously leveraging opportunities to work across and services and systems, addressing the holistic needs of people.

The strategies promote the delivery of HIV, viral hepatitis and STI services through implementing a PHC approach, including through decentralized and community-based service delivery when doing so makes sense. They also recognize an increasingly important role for the private sector.

And they promote a PHC approach that leverages empowered community engagement and multisectoral policy and actions to address determinants of health and ensure that services needed throughout the life-course are delivered as close as feasible to people's homes.

Importantly, the strategies note that importance of giving priority to the needs of people living with and most severely affected by HIV, viral hepatitis and STIs – recognizing that stigma and discrimination in health settings continue to create barriers to access.

Key challenges include an acute decline in disease-specific funding, especially in low- and middle-income countries, and growing reliance on domestic resources and increasingly competitive bilateral and multilateral sources of support. This situation became more acute with the drastic reduction of United States Government support for global health in 2025 – with HIV and other global programmes, generously championed by the United States for several decades, among areas most severely affected *(2)*.

As programmes and services transition to new models and ways of working, providing adequate support and capacity building to health-care personnel, including formal and informal community health workers, is critical to maintaining service quality and the confidence of service users.

Simply put, the PHC approach must be made to work for people living with and affected by HIV and related health problems, including key, marginalized and underserved populations while addressing the needs of the entire population.

Progress in 2024 noted the importance of PHC

In 2024, WHO outlined the first two years of strategy implementation in *Implementing the global health sector strategies on HIV, viral hepatitis and sexually transmitted infections, 2022–2030: report on progress and gaps 2024 (3)*.

The report reflected on how countries responded to the challenge to accelerate work to integrate services and systems and to strengthen work with communities using a PHC approach. Many countries gave priority to approaches to address common comorbidities. For example, in 2022, in 97 countries, at least 90% of people diagnosed with tuberculosis (TB) knew their HIV status. Data on service integration collected through the Global Hepatitis Reporting System show that, of the 27 WHO focus countries reporting this information, 78% reported that hepatitis B testing and treatment services are integrated with existing services in HIV and antiretroviral therapy clinics and/or pre-exposure prophylaxis (PrEP) sites, and about 63% reported that these are integrated into PHC. For hepatitis C, 70% of countries reported that services are integrated with existing services in HIV and antiretroviral therapy clinics and/or PrEP sites, and 67% reported that these are integrated into PHC.

The progress report also described gaps and missed opportunities: for example, access to essential health commodities remains inadequate in many low- and middle-income countries in PHC. In 2023, most WHO focus countries for the viral hepatitis response had included tenofovir disoproxil fumarate for hepatitis B in their national viral hepatitis treatment guidelines and in national essential medicines lists; yet only 45% of reporting countries have it available for use in PHC. Similarly, most reporting WHO focus countries had included sofosbuvir and daclatasvir for hepatitis C in their national treatment guidelines and essential medicines lists, yet only 30% reported the availability of these medicines in PHC.

The PHC approach offers the most promising and powerful way to tackle HIV, viral hepatitis and STIs while meeting the wider needs of populations. It enables integrated health services – spanning health promotion, disease prevention, treatment, rehabilitation and palliation – to be implemented as close as possible to where people need them, emphasizing the importance of public health functions. The PHC approach's second compelling aspect is its decided orientation towards multisectorality and disease prevention, which is critical for communicable disease control. Further, the PHC approach not only brings essential disease prevention and care closer to the people but puts them directly in charge of it through community engagement and empowerment. This approach is even more important in 2025 than it has ever been as the world comes to terms with a rapidly evolving global health landscape.

Low- and middle-income countries are increasingly committed to and are shifting from fragmented, often donor-supported and disease-focused and provider-centred care, to integrated health systems and PHC approaches. In many contexts, this transition is helping to enhance health care quality, accessibility and sustainability, reducing new infections and improving long-term health outcomes. Integrated care rooted in PHC ensures that services are accepted by communities, economically viable and able to produce lasting benefits for individuals and society.

Rationale

This policy brief seeks to document and highlight progress and challenges in converging, linking and integrating HIV, viral hepatitis and STI services with PHC in several low- and middle-income countries.

The primary objective is to present the experiences of selected low- and middle-income countries in this integration process, providing insight for other countries, policy-makers within the health sector and non-heath sectors, communities and civil society organizations and multilateral and funding partners.

The brief includes examples of countries evolving their health-care services and systems from donor-funded approaches to more domestically funded and managed responses, offering guidance and lessons learned on integrating HIV, viral hepatitis and STI services with PHC.

The brief draws on two sources. Section One describes and summarizes examples that have been discussed at WHO co-hosted conference sessions and webinars while also drawing on examples from countries that have engaged in the UNAIDS-led process to develop country HIV sustainability roadmaps. Some of these reference the WHO/UNICEF Operational Framework for Primary Health Care *(4) (*see Fig. 1), which has been used to identify convergent actions and opportunities to jointly strengthen PHC and advance disease-specific goals. The Operational Framework describes four strategic and 10 operational levers to ensure a comprehensive PHC approach.

This is then followed by Section Two, which provides more detailed insight from four countries: Ethiopia, Indonesia, Rwanda and Zambia.

Successful integration occurs when countries focus on several strategic and operational levers of PHC at the same time. The examples and case studies demonstrate the importance of addressing these interdependent strategic and operational levers simultaneously, with all examples noting the importance of political will and commitment as a prerequisite to success.

Development of the policy brief

In 2023 the WHO Global HIV, viral hepatitis and sexually transmitted infections programmes department developed a protocol for the development of case studies on the integration of HIV, viral hepatitis and STI services with PHC. The case study protocol outlined a systematic approach to document and analyze the processes, outcomes, and lessons from various countries implementing integration efforts.

The development of the policy brief was coordinated by a WHO internal steering committee comprising staff from WHO headquarters, and regional and country offices working on HIV, viral hepatitis, STIs, PHC, quality of care and TB.

Staff members from PATH and the Global Network of People Living with HIV also informed the development of the policy brief in their capacity as organizations in official relations with WHO and as part of their agreed workplans.

The topic of this policy brief was presented to representatives from countries and communities most impacted by HIV, viral hepatitis and STIs at the 25th International AIDS Conference in July 2024 and was informed by a webinar co-hosted by PATH and WHO on considerations for integrating HIV and strengthened PHC systems held in October 2024. During webinar discussion key messages and themes emerged that have informed this policy brief.

Preliminary findings and an outline of the policy brief were presented to the Strategic and Technical Advisory Group on HIV, Viral Hepatitis and Sexually Transmitted Infections (STAG-HHS) in November 2024 which subsequently endorsed the approach and encouraged broad dissemination of findings.

The final draft of this document underwent peer review by external experts from Ministries of Health and UN entities.

As part of the process all external experts submitted to WHO a declaration of interest disclosing potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence in relation to the subject matter of this guidance. WHO reviewed each of the declarations and concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subjects discussed at the meeting or covered by the guidance.

The introduction to Section Two of this policy brief provides more information about the methodology used to collect data for the four country case studies.

Fig. 1. **The WHO/UNICEF Operational Framework for Primary Health Care**



Source: WHO & UNICEF (4).

Section One



The Operational Framework for Primary Health Care describes how the core elements of the PHC approach can be translated into results through 14 interrelated levers for action and investment. The strategic levers are:

- political commitment and leadership
- governance and policy frameworks
- funding and allocation of resources
- engagement of the community and other stakeholders.

The operational levers are:

- models of care
- PHC workforce
- physical infrastructure
- medicines and other health products
- engagement with private-sector providers
- purchasing and payment systems
- digital technologies for health
- systems for improving the quality of care
- PHC-oriented research
- monitoring and evaluation.

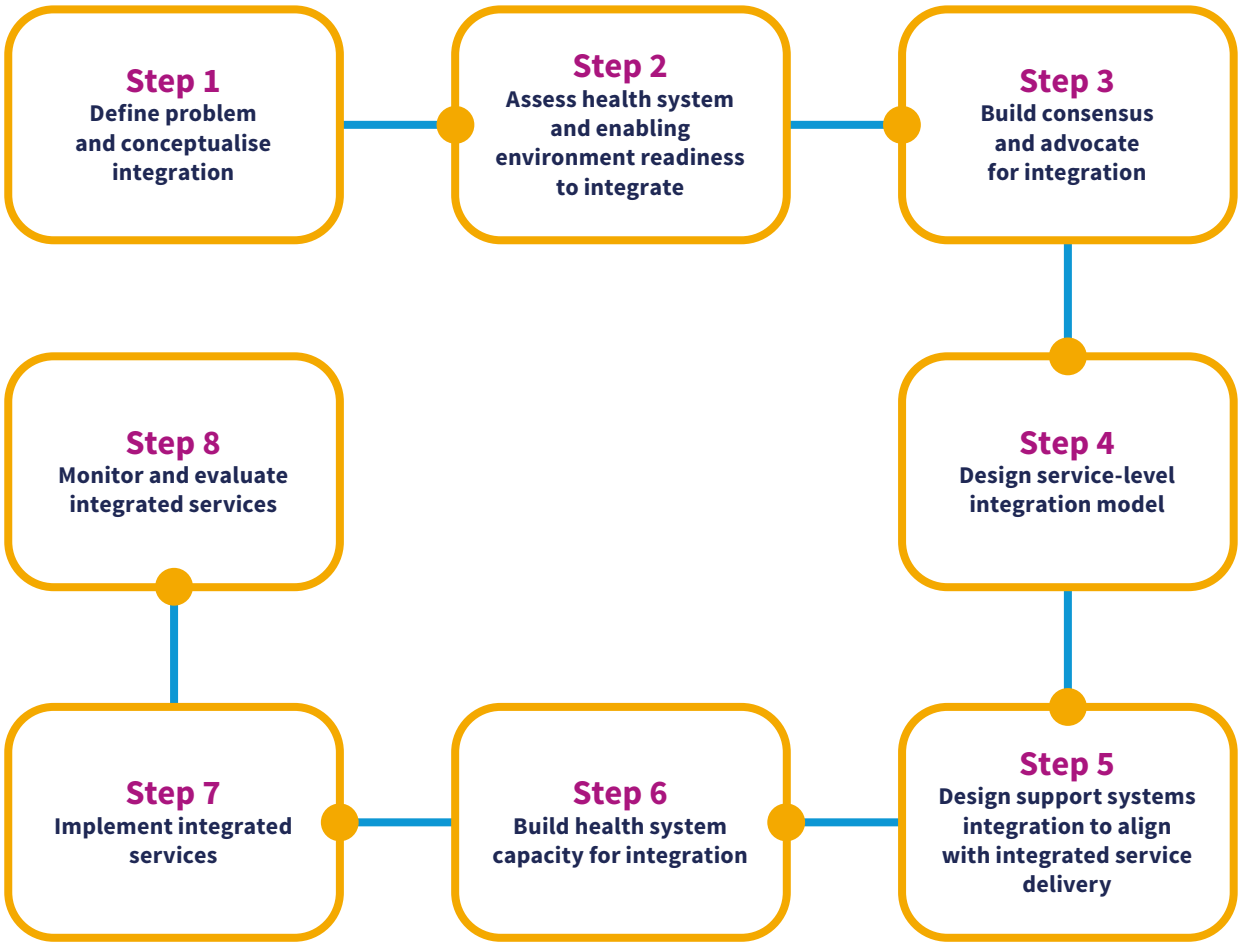
The PHC levers serve as a useful framework for identifying and leveraging opportunities to jointly strengthen PHC and meet disease-specific goals.

The examples and case studies show how combining different strategic and operational levers can jointly advance PHC and disease-specific goals.

PHC and disease-specific stakeholders should ideally work together to decide and plan for which levers should be given priority in their specific national and subnational settings and should define the collaborative or synergistic actions to be carried out in association with these levers. Various guidance documents help to outline the key steps required to support this.

In 2024, Genesis Analytics developed a guide on integrating HIV programmes and services into PHC *(5)*. It outlines a generic yet adaptable pathway for integrating HIV services into PHC, offering detailed considerations for each step of the process. Fig. 2 outlines the key steps.

Fig. 2. **Generic HIV programme integration pathway**



Source: adapted from Integrating HIV programs into primary health care systems: practical considerations for country decision-makers (5).

The strategic and operational levers are referenced throughout this publication to highlight which combination of levers has been implemented in the examples and case studies described. The country examples and case studies demonstrate the importance of addressing several strategic and operational levers simultaneously, with all examples noting the importance of political will and commitment as a prerequisite to success.

Section One summarizes examples from work undertaken between WHO and partners in 2023–2024 – the period before the 2025 seismic shifts in global health financing and political support.

The first series of examples emerged from a 2024 independent evaluation of the contribution of the UNAIDS Joint Programme to strengthening HIV and PHC outcomes *(6)*. Although the evaluation explored the role of the UNAIDS Joint Programme in supporting PHC in countries, it also noted some global themes, including a lack of clarity about the extent to which HIV investments, infrastructure, capacity and systems established for the HIV response have strengthened wider health systems. It noted several areas from HIV and PHC integration that could be adapted and applied more widely to benefit other health areas and further the PHC approach. These included approaches that benefited both PHC and HIV goals in the following areas: differentiated service delivery; person-centred strategic information; use of digital technology and virtual approaches; multisectoral action; community-based and community-led interventions; and strategies for reaching marginalized and vulnerable populations.

Several country case studies were also commissioned as part of the UNAIDS Joint Programme evaluation and are summarized here. Each case study explores areas of best practice and areas for priority focus.

A linked independent evaluation of the WHO Special Programme on Primary Health Care *(7)* also considered barriers and enablers to closer working and convergence between HIV and PHC, Fig. 3 describes the barriers and enablers for closer working identified in the second evaluation.

Fig. 3. **Barriers and enablers to closer working and convergence between HIV and PHC**

Global barriers

Lack of globally agreed core indicators on HIV and PHC integration and interlinkage

Lack of global guidance on what services to integrate and how and on the appropriate configuration of integrated and specialized services in different contexts

Weak leadership and understanding on HIV and PHC integration – concepts, strategy and action

Global enablers

Existence of normative and technical guidance for greater integration of services

Intersecting barriers

Lack of political commitment and funding for PHC and competition between United Nations agencies for resources and turf issues

Concerns about loss of funding among government and civil society organization HIV stakeholders

Limited donor or country incentives to give priority to integrating HIV and PHC

Concerns about reversing gains, compromising the quality of HIV care and HIV data and excluding key populations

Intersecting enablers

The COVID-19 pandemic facilitated innovation and potential for more integrated approaches

Global health strategies and national health sector and disease strategies support the integration of services

Country barriers

Underfunding and weak capacity of primary care; investment skewed towards tertiary levels rather than primary care facilities and services

Primary care not client-centred, deterring access

Shortages of human resources for health in PHC; lack of HIV knowledge and skills of primary care workers in settings with low HIV prevalence; costs associated with training, sensitizing and supervising primary care workers

Country enablers

Infrastructure exists to integrate more services

Source: Evaluation of the WHO Special Programme on Primary Health Care: report (7).

Angola

The 2024 UNAIDS Joint Programme PHC and HIV evaluation in Angola *(8)* identified significant advances in expanding access to HIV treatment and prevention services (Fig. 4) but also highlighted ongoing challenges that hinder the attainment of universal health coverage and greater integration and convergence with PHC.

The opportunities for further HIV and PHC integration and convergence are significant but require systems strengthening and structural change within the Ministry of Health and the National Institute for the Fight Against AIDS, with strengthened linkage to social protection and a greater focus on strengthening health workforce capacity.

The lack of an organized community health strategy has hampered a meaningful scaled contribution of communities, civil society organizations and key populations to both the HIV response and to advancing the PHC approach.

Further work is required to support the government to apply key lessons learned from the HIV response to PHC strengthening approaches given the higher levels of donor investment in the former.

Although Angola has implemented initiatives to improve HIV services, these efforts continue to be impeded by weak broader health systems. Health-care facilities in Angola require significant investment and improvement to adequately support the delivery of high-quality HIV and related services. This includes not only building new facilities but also upgrading existing ones and ensuring that they are equipped with the necessary medical supplies and technology. In addition, the workforce needs to be better trained and supported to manage the complexities of delivering high-quality HIV care. Community health workers play a crucial role in bridging gaps between the health-care system and the population, especially in rural and underserved areas. Affected communities can be more systematically engaged in decision-making for both disease-specific and wider health issues.

Data collection and analysis are emphasized as needing further enhancement. The ability to track progress and identify gaps in service provision is fundamental for informed decision-making and strategic planning. Greater investment in data systems to monitor the effectiveness of HIV interventions is needed in Angola.

Although Angola has made important advances in addressing HIV through PHC, the evaluation identifies critical areas for improvement. Strengthening health systems, investing in infrastructure, enhancing workforce training and addressing social determinants are essential steps toward achieving better health outcomes and universal health coverage.

PHC levers identified for priority action in Angola to advance HIV and PHC common goals

PHC levers in Angola	
Demonstrates evidence of political commitment and leadership	
Governance and policy frameworks are in place	
Demonstrates evidence of funding and allocation of resources	
Engagement of community and other stakeholders is referenced	
Models of care are adapted	
PHC workforce is strengthened	
Physical infrastructure is adapted or improved	
Includes a focus on strengthening access to medicines and other health products	
Evidence of engagement with private-sector providers or partners	
Purchasing and payment systems are adapted	
Includes a focus on digital technologies for health	
Systems are in place for improving the quality of care	
PHC-oriented research is included	
Monitoring and evaluation plans in place	

Context-specific innovations and lessons learned

Evidence of a mature system or an approach with lessons that can be readily applied to other country contexts

Botswana

The 2024 UNAIDS Joint Programme evaluation of PHC and HIV in Botswana *(9)* recognizes a conducive policy environment for integrating HIV services with other services within primary care, and there appears to be a shared appreciation that this evolution is needed. Although progress is being made in scaling up integration at the service delivery level, little attention has been focused on integrating the systems that support the services.

Botswana is recognized for its leadership and commitment to tackling HIV, boasting one of the highest coverage rates for antiretroviral therapy in the world. The evaluation highlights that the government has made substantial investments in health-care infrastructure and human resources, which has contributed to improved access to HIV services. The integration of HIV care within the PHC framework is seen as a critical strategy and political priority for ensuring that individuals receive comprehensive health-care services.

Community engagement in delivering HIV services is a central part of Botswana's success and is helping to advance important elements of the PHC approach. Community health workers and peer educators play a crucial role in reaching underserved populations, providing education and facilitating access to testing and treatment. The evaluation emphasizes that community involvement not only improves health outcomes but also fosters trust and reduces the stigma associated with HIV.

Despite these successes, the evaluation identifies several challenges in the Botswana health system. A key issue is the variability in the quality of health service delivery across regions, especially between urban and rural areas. Urban centres have access to better resources and facilities, whereas rural communities often face barriers such as limited health-care personnel and inadequate infrastructure. This disparity highlights the need for targeted interventions to ensure equitable access to HIV services throughout the country.

The evaluation also underscores the necessity of addressing health system barriers, including supply chain issues and the availability of essential medications. Ensuring a consistent supply of antiretroviral therapy and other critical medications is vital to maintaining treatment adherence and improving health outcomes. Strengthening the logistics and supply chain management systems is therefore essential for effective service delivery for both diseases and wider health issues.

Data management and monitoring are identified as crucial areas for improvement. The evaluation stresses the need for more robust data systems to track progress in HIV treatment and prevention efforts within the context of wider efforts of health management and information systems.

Botswana has committed to enhancing community engagement, ensuring equitable service delivery, strengthening supply chains, improving data management and addressing social determinants of health as part of its response to HIV and towards convergent PHC strengthening goals. Fig. 5 demonstrates where Botswana is seeking to leverage several strategic and operational levers as part of its approach to reach HIV and PHC goals.

PHC levers identified for priority action in Botswana to advance HIV and PHC common goals

PHC levers in Botswana	
Demonstrates evidence of political commitment and leadership	
Governance and policy frameworks are in place	
Demonstrates evidence of funding and allocation of resources	
Engagement of community and other stakeholders is referenced	
Models of care are adapted	
PHC workforce is strengthened	
Physical infrastructure is adapted or improved	
Includes a focus on strengthening access to medicines and other health products	
Evidence of engagement with private-sector providers or partners	
Purchasing and payment systems are adapted	
Includes a focus on digital technologies for health	
Systems are in place for improving the quality of care	
PHC-oriented research is included	
Monitoring and evaluation plans in place	

Context-specific innovations and lessons learned

Evidence of a mature system or an approach with lessons that can be readily applied to other country contexts

Pakistan

The 2024 UNAIDS Joint Programme evaluation of PHC and HIV in Pakistan *(10)* highlights areas of progress and significant challenges including in combating HIV and strengthening PHC.

The HIV response in Pakistan is still largely verticalized in practice, with standalone clinics for antiretroviral therapy and services for preventing vertical transmission being given priority and with separate financing, separate data and procurement systems and separate funding streams and management systems for HIV.

There is a plan to develop a clear and comprehensive strategy building on the PHC approach of the national health vision, learning from the PHC model of care applied in two districts and assessing the potential for increased integrating service delivery and patient-centred care for marginalized populations at risk of various infections and diseases (HIV, hepatitis, TB, STIs, mental health disorders, drug use etc.).

Although some progress has been made, several challenges hinder the integration of HIV into PHC in Pakistan such as financial constraints, stigma, discrimination, limited awareness and fragmented data management systems. On a positive note, a well-established network of PHC facilities and community health workers and the integration of HIV interventions into the universal health coverage benefit package provide enablers for a more PHC-aligned HIV response. Overcoming these barriers and leveraging the identified enablers are pivotal to advancing the integration of HIV into PHC, ensuring comprehensive and accessible health care for all.

Pakistan has faced considerable barriers in the HIV response, including stigma, discrimination and lack of HIV awareness. In response, the government, along with affected communities and other stakeholders, has initiated programmes aimed at increasing access to HIV testing, treatment and prevention services. The integration of HIV services within a PHC framework is identified as a key strategy to enhance the accessibility and efficiency of health-care delivery.

Community health workers and peer educators have proven effective in reaching key and marginalized populations, including men who have sex with men, sex workers and injecting drug users. Community-led initiatives significantly improve awareness, reduce stigma and encourage individuals to seek HIV testing and treatment.

Access to HIV services is often limited in rural areas, where health-care infrastructure is inadequate and trained personnel are scarce. The evaluation calls for targeted

interventions to ensure that vulnerable populations in these areas receive the necessary support and services.

The evaluation also highlights the need for improved coordination among various stakeholders involved in HIV prevention and treatment, including the private sector. Although multiple organizations are working to address HIV in Pakistan, a lack of clear collaboration and governance mechanisms can lead to inefficiency and gaps in service delivery. Strengthening partnerships between government agencies, nongovernmental organizations and community groups is essential for creating a more cohesive response to the HIV epidemic.

Data management is another critical area identified in the evaluation. Accurate data collection and analysis are imperative for understanding the scope of the HIV epidemic and for monitoring the effectiveness of interventions. The report emphasizes the need for robust health information systems that can provide timely data to inform policy decisions and programme planning.

Fig. 6 shows how Pakistan has identified the need to act within the domains of several strategic and operational levers as part of its approach to reach both HIV and PHC goals.

PHC levers identified for priority action in Pakistan to advance HIV and PHC common goals

PHC levers in Pakistan	
Demonstrates evidence of political commitment and leadership	
Governance and policy frameworks are in place	
Demonstrates evidence of funding and allocation of resources	
Engagement of community and other stakeholders is referenced	
Models of care are adapted	
PHC workforce is strengthened	
Physical infrastructure is adapted or improved	
Includes a focus on strengthening access to medicines and other health products	
Evidence of engagement with private-sector providers or partners	
Purchasing and payment systems are adapted	
Includes a focus on digital technologies for health	
Systems are in place for improving the quality of care	
PHC-oriented research is included	
Monitoring and evaluation plans in place	

Context-specific innovations and lessons learned

Evidence of a mature system or an approach with lessons that can be readily applied to other country contexts

Brazil

In Brazil, 70% of the population has access to the Government's PHC flagship initiative, the Family Health Strategy (Estratégia de Saúde da Família) *(11)*.

The Municipality of Abaetetuba-PA has 156 000 residents, and only 53% are covered with family health care *(11)*. Ten years ago, residents were facing significant health challenges, including high mortality rates for women of childbearing age and many teenage pregnancies and cases of syphilis, HIV and viral hepatitis. The population clearly needed better sexual and reproductive health-care services. A project was designed with the community and involved health interventions with adolescents, adults and older people, addressing sexuality, appreciation and respect.

Young people and adolescents took part in art workshops, personal and social education activities, theatre and dance and addressed issues such as sexual diversity, bullying, homophobia, citizenship and culture of peace. In addition, all health-care services in the municipality worked in partnership to focus on care and providing high-quality services.

There was rapid noticeable improvement, with 7028 prenatal care visits in 2018 versus 2862 in 2016; 4161 sexual and reproductive health-care service visits versus 2507 visits in 2016; 1291 rapid tests for hepatitis B, syphilis and HIV in 2017 versus 432 tests in 2016; and 1880 cervical cancer screenings in 2018 versus 143 in 2016 *(11)*.

A separate study *(12)* assessed the impact of the Brazilian Family Health Strategy, one of the largest PHC programmes in the world, on HIV-related incidence and mortality using a cohort of 3.4 million individuals over a nine-year study period.

HIV-related data from individuals aged 13 years or older who were members of a nationwide cohort of the poorest Brazilian people (the 100 Million Brazilians Cohort) were analysed from 1 January 2007 to 31 December 2015, and residents in municipalities with no coverage were compared with residents in municipalities with full Brazilian Family Health Strategy coverage.

Brazilian Family Health Strategy coverage was associated with lower HIV incidence and mortality, with the impact on incidence greater among females and the black community. The effect of Brazilian Family Health Strategy coverage on mortality was greater among males. For both outcomes, the effect of Brazilian Family Health Strategy coverage was stronger among people 35 years or older (0.62, 95% confidence interval 0.53–0.72 for incidence and 0.56, 95% confidence interval 0.43–0.72 for mortality), although people 13–34 years old had lower incidence (0.83, 95% confidence interval 0.72–0.96) *(12)*.

The study concluded that Brazilian Family Health Strategy coverage affects HIV-related morbidity and mortality among the most vulnerable populations in Brazil, thus clearly emphasizing that the PHC approach offers the most promising and powerful way to tackle HIV, viral hepatitis and STIs while meeting the wider needs of populations.

Fig. 7 shows examples from the studies in Brazil. The Brazil experience highlights how strong political commitment combined with a clear system for service delivery and community engagement can secure results – including for the most vulnerable people in society.

PHC levers identified for priority action in Brazil to advance HIV, sexual and reproductive health, viral hepatitis and STI goals with PHC goals

PHC levers in Brazil	
Demonstrates evidence of political commitment and leadership	
Governance and policy frameworks are in place	
Demonstrates evidence of funding and allocation of resources	
Engagement of community and other stakeholders is referenced	
Models of care are adapted	
PHC workforce is strengthened	
Physical infrastructure is adapted or improved	
Includes a focus on strengthening access to medicines and other health products	
Evidence of engagement with private-sector providers or partners	
Purchasing and payment systems are adapted	
Includes a focus on digital technologies for health	
Systems are in place for improving the quality of care	
PHC-oriented research is included	
Monitoring and evaluation plans in place	

Context-specific innovations and lessons learned

Evidence of a mature system or an approach with lessons that can be readily applied to other country contexts

The next two country examples emerged from work undertaken by PATH's PHC programme *(13)*. WHO and PATH have strengthened collaboration in recent years in support of strengthening disease-specific outcomes and PHC – including by focusing on integration and sustainability.

Integrating HIV and noncommunicable disease services by strengthening PHC in Kenya

Kenya has made significant strides towards controlling the HIV epidemic, with the number of people newly infected with HIV declining by 83% and the number of people dying from HIV-related causes declining by 65% between 2013 and 2022 *(14)*. Nevertheless, services for people living with and vulnerable to HIV were largely standalone and delivered through specialized HIV clinics, creating gaps in access to comprehensive health care, including for noncommunicable diseases and mental health.

Beginning in 2021, the Government of Kenya called for integrated delivery of HIV with other health services within Kenya's 2021–2025 AIDS Strategic Framework II. The Government of Kenya's 2023 health reforms further underscored this push towards integrated health services delivered through the PHC platform as the vehicle driving Kenya's Vision 2030 strategy for sustainable universal health coverage.

In 2024, the Ministry of Health developed a framework for integrating HIV services as part of PHC. Following this commitment to integrated services delivered through PHC, the Government of Kenya convened partners to test integrated HIV delivery models, ranging from expanding routine HIV and STI screening within maternal and child health clinics to offering bundled HIV and noncommunicable disease screening and care.

These delivery models included integrated HIV, noncommunicable disease and PHC services at both facility and community entry points:

- offering screening and referrals to follow-up care for hypertension, diabetes, cervical cancer and mental health conditions during clinical check-ups at comprehensive HIV care clinics;
- integrating HIV and noncommunicable disease screening and referrals for follow-up care into PHC outpatient services and community screening efforts; and
- delivering bundled treatment and adherence support for HIV and/or noncommunicable diseases through differentiated care models (including facility fast-track medication pick-up circuits), community support groups and self-care packaging solutions (such as Diabetes CarePak).

Efforts to integrate HIV and noncommunicable diseases have also extended to systems-level components, including:

- data systems: designing and deploying a noncommunicable disease–specific module within the expanded Kenya Electronic Medical Record system to integrate client and drug-dispensing data on diabetes and hypertension alongside HIV;
- supply chain management: providing affordable medications to individuals living with HIV and noncommunicable diseases by sharing information on and encouraging registration for Kenya's Social Health Insurance Fund; and
- health workforce: training HIV and PHC clinical and community staff to provide HIV and noncommunicable disease screening, referral and care services while continuing supportive supervision, on-the-job mentorship and community-led monitoring to ensure high-quality provision of newly integrated health services.

Integrated HIV, noncommunicable disease and PHC models demonstrated successful case finding and linkage to care in Kenya (Fig. 8). In four counties of western Kenya alone, more than 83 010 people living with HIV were screened for hypertension, with 8020 people living with HIV and hypertension receiving affordable and convenient care to manage both HIV and noncommunicable diseases as of December 2024 *(15)*. Focusing on noncommunicable disease screening and care while maintaining suppressed viral loads among people living with HIV is critical for supporting longevity and maintaining quality of life in this integrated approach. These efforts are now being integrated into Kenya's primary care networks, where PATH is partnering with the National AIDs & STI Control Programme and Nyamira County to test a country peer-learning approach of integrating HIV services into primary care networks and among community health posts and community health promoters, including digitalizing HIV community reporting through the national electronic Community Health Information System, improving community HIV commodity supply chains and training, equipping and facilitating supportive supervision. Learning from this approach will be used to scale up embedding of HIV services within primary care networks by December 2025.

Kenya's models for integrating HIV and noncommunicable diseases considered most of the strategic and operational levers of the Operational Framework for Primary Health Care *(4)*.

PHC levers identified for priority action in Kenya to advance HIV and noncommunicable disease integration and PHC common goals

PHC levers in Kenya	
Demonstrates evidence of political commitment and leadership	
Governance and policy frameworks are in place	
Demonstrates evidence of funding and allocation of resources	
Engagement of community and other stakeholders is referenced	
Models of care are adapted	
PHC workforce is strengthened	
Physical infrastructure is adapted or improved	
Includes a focus on strengthening access to medicines and other health products	
Evidence of engagement with private-sector providers or partners	
Purchasing and payment systems are adapted	
Includes a focus on digital technologies for health	
Systems are in place for improving the quality of care	
PHC-oriented research is included	
Monitoring and evaluation plans in place	

Context-specific innovations and lessons learned

Evidence of a mature system or an approach with lessons that can be readily applied to other country contexts

Community-led design and delivery of integrated HIV and PHC services in Viet Nam

Until 2014, most of Viet Nam's HIV services were delivered in the public sector in partnership with key population–led community organizations. A 2014 change in the enterprise law enabled key population organizations to register as social enterprises and to establish their own community health-care clinics.

Over the past decade, key population–led and –owned clinics have opened with approvals and oversight from local health departments and the Ministry of Health. More than 25 such clinics and pharmacies exist nationwide and are concentrated in urban communities and ones with and higher HIV prevalence. These clinics have evolved their services, starting first with those focused on HIV testing, antiretroviral therapy and PrEP (from 2017 on) *(16, 17)* and then expanding incrementally to more holistic care, including for viral hepatitis *(18)*, STIs, noncommunicable diseases, dental care, skin diseases, mental health counselling, addiction management and transgender health care *(19)*.

An initiative of the United States Agency for International Development and PEPFAR partnered with 15 of these clinics to strengthen their quality of care and business models to optimize their sustainability chances without donor funding. From October 2020 to 2024, these integrated clinics served more than 37 400 clients. Most clients first visited seeking PrEP or HIV testing, with more than half obtaining additional services at the clinic: 75% received STI testing services, 43% were screened for hepatitis B, 48% were screened for hepatitis C and 51% received mental health counselling; 26% of the new PrEP clients enrolled in PrEP after first seeking other PHC services *(20)*.

More than 3000 individuals sought antiretroviral therapy at these clinics. Mental health screening and care resulted in increased antiretroviral therapy and PrEP retention. The approach also resulted in an increase in transgender women seeking health care, including PrEP services. These clinics received small grants from PEPFAR and the Global Fund to offer free PrEP services (about half of the PrEP clients in Viet Nam sought PrEP from key population–led or private-sector clinics).

In an assessment conducted after United States Government funding for these clinics ended (May 2025), 24% are fully operating on commercial revenue. They have addressed financial losses from the United States Government by drawing on profits to sustain and right-size harder-hit clinics when needed. A total of 46% of the clinics are sustained through a blend of commercial income (50–70%) and grants other than the United States Government. Some of these clinics reduced workforce and others transferred staff to other (non-health) businesses to navigate abrupt funding losses. A total of 30% of the clinics, which represent the newest (less than two years of operation), remain open but had not yet established an adequate client base to sustain their services through commercial revenue. Although they are operational now, these clinics risk closing over the months to come.

Fig. 9 illustrates which of the strategic and operational levers of the Operational Framework for Primary Health Care were important to the integration work in Viet Nam.

PHC levers identified for priority action in Viet Nam, which placed communities at the heart of its work to design, develop and deliver comprehensive services

PHC levers in Viet Nam	
Demonstrates evidence of political commitment and leadership	
Governance and policy frameworks are in place	
Demonstrates evidence of funding and allocation of resources	
Engagement of community and other stakeholders is referenced	
Models of care are adapted	
PHC workforce is strengthened	
Physical infrastructure is adapted or improved	
Includes a focus on strengthening access to medicines and other health products	
Evidence of engagement with private-sector providers or partners	
Purchasing and payment systems are adapted	
Includes a focus on digital technologies for health	
Systems are in place for improving the quality of care	
PHC-oriented research is included	
Monitoring and evaluation plans in place	

Context-specific innovations and lessons learned

Evidence of a mature system or an approach with lessons that can be readily applied to other country contexts

Section Two



Approach and methods

This part of the policy brief used a qualitative case study approach to examine how HIV, viral hepatitis and STI services were integrated with PHC approaches in four countries: Ethiopia, Indonesia, Rwanda and Zambia.

Data were collected in 2024 through document reviews, interviews with health-care providers, stakeholders responsible for coordinating programmes, communities and development partners.

Before the interviews, the WHO writing team completed literature searches and reviewed programme strategy documents, programme data, donor proposals and results reports.

Verbal informed consent was obtained from all participants before conducting interviews. Participants were informed about the study's purpose, procedures, confidentiality, non-attribution, and their right to withdraw at any time without any consequences.

Data analysis applied a detailed and systematic approach to examine the information collected from documents, interviews, and other sources. Key themes, patterns, and insights related to the integration of HIV, Hepatitis, and STI services with primary healthcare (PHC) were identified.

Analysis was guided by a CARE framework approach (Context, Action, Results, Evaluation):

- Context: Environmental, social, economic, and political factors influencing integration.
- Action: Specific strategies and activities implemented during the integration.
- Results: Outcomes and outputs resulting from the integration.
- Evaluation: Effectiveness of the actions and understanding of the outcomes.

Note that the results and insights were generated for the four countries before the significant disruptions to development assistance in the first quarter of 2025. Nevertheless, the results are perhaps even more important precisely because the world is entering a new era of global health and development assistance, which necessitates a specific focus on PHC-oriented health systems that can meet the health needs of populations.

Results

In an era marked by global health challenges, disease integration has emerged as a crucial strategy to enhance health-care delivery and improve health outcomes. Integrating services and programmes for HIV, viral hepatitis, TB, sexual and reproductive health including STIs, and noncommunicable diseases and mental health within PHC-oriented health systems is vital to achieving universal health coverage. The PHC approach ensures comprehensive people-centred care and efficient use of resources and addresses multiple health problems in a cohesive manner.

The case studies from Ethiopia, Indonesia, Rwanda and Zambia offer valuable insight into the implementation of disease control and integration in the context of PHC. They underscore the global need to adopt this strategy, demonstrating a commitment to integrating multiple disease services within PHC. These case studies highlight common strategies, unique strengths and areas of difference, providing important insight into how integration and convergence can be effectively achieved.

Common themes emerged from country experiences

Political commitment and leadership

All countries emphasize strong political commitment and leadership. High-level political endorsement drives initiatives and aligns them with national priorities. In Rwanda, political commitment from the highest levels facilitated hepatitis C elimination; in Ethiopia, political commitment and leadership have been crucial, with supportive policies and frameworks enhancing integration efforts with community-based organizations and development partners.

Task sharing and workforce development

Task sharing is a pivotal strategy used across these countries to optimize health-care resources and extend service reach. In Ethiopia, health extension workers are trained to provide basic services, and Zambia uses community health workers to deliver integrated care. Indonesia also uses task sharing to address resource constraints and ensure comprehensive care. All four countries have used task sharing to extend health care reach and optimize resource use.

Community engagement

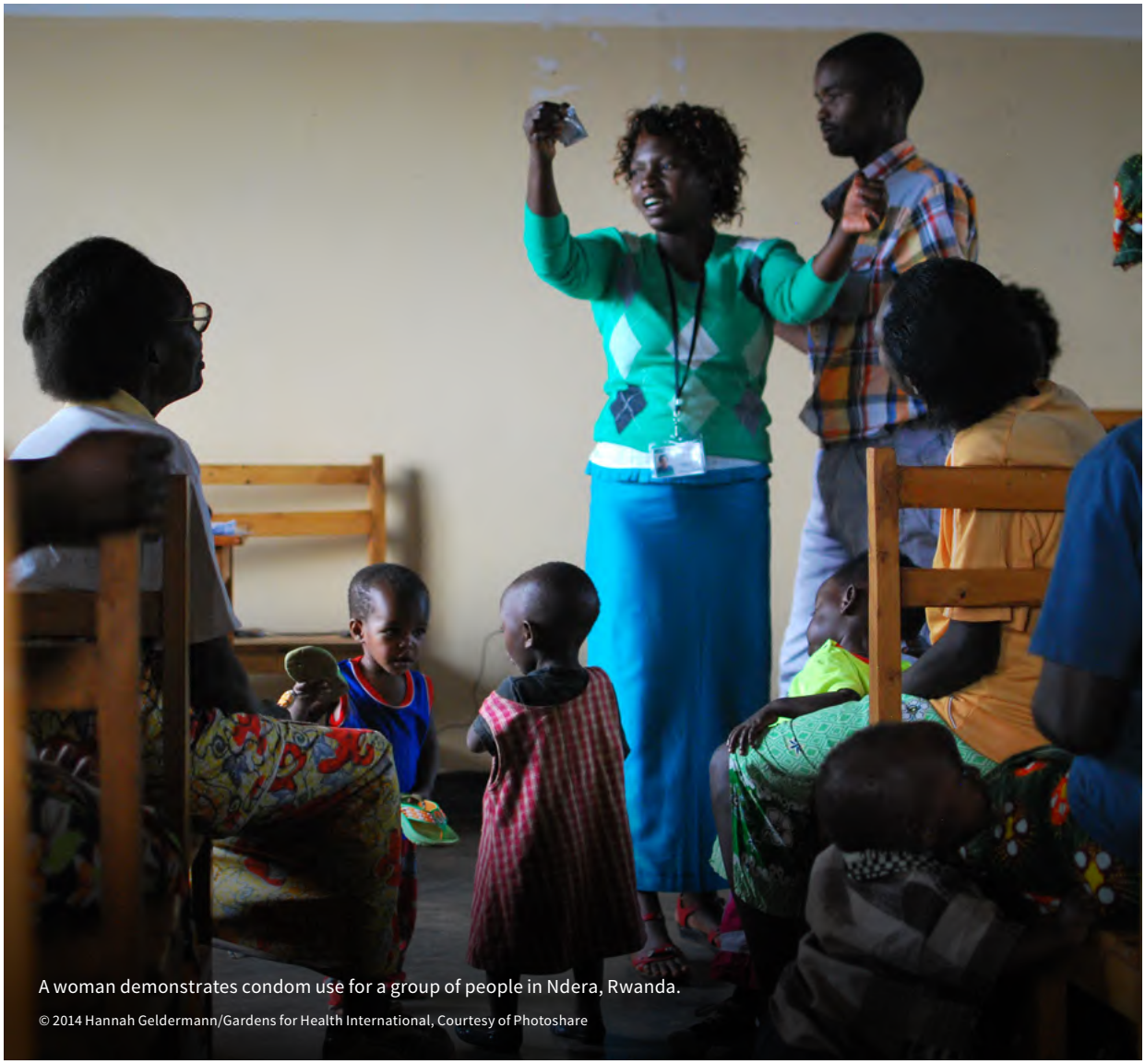
Engaging communities in planning and implementation is essential for successfully integrating disease-specific services and programmes into broader systems and PHC. Rwanda's community engagement in hepatitis C activities has improved service uptake. Ethiopia's Health Extension Program actively involves community health workers, enhancing service delivery.

Innovative financing mechanisms

Securing adequate and sustainable financing is critical. Efforts to secure adequate funding, negotiate reduced costs for treatment and implement health insurance schemes are essential for sustaining integration activities into systems and services. Rwanda negotiated reduced costs for hepatitis C treatment, Indonesia used community-based health insurance schemes and Zambia implemented the National Health Insurance Scheme to provide universal health coverage.

Integrated service delivery models and decentralized service delivery

One-stop clinics and decentralized services are common across all four countries. Rwanda's decentralized treatment model for hepatitis C, Indonesia's integrated PHC approach, Ethiopia's one-stop shopping for HIV, TB, STI and hepatitis services and Zambia's integrated HIV and noncommunicable disease services all exemplify this strategy. Providing services at PHC centres and community levels helps to improve accessibility and reduce out-of-pocket costs.



A woman demonstrates condom use for a group of people in Ndera, Rwanda.

© 2014 Hannah Geldermann/Gardens for Health International, Courtesy of Photoshare

Unique strengths and best practices

Ethiopia

Ethiopia has made significant strides in integrating HIV, hepatitis, TB and STI services through its PHC system. Initiatives like the Health Extension Program and community-based health insurance schemes have strengthened community engagement and decentralized service delivery. Ethiopia's success is underpinned by strong political commitment and coordinated governance.

Indonesia

Indonesia's efforts to integrate HIV, hepatitis, TB and STI services into PHC are notable for their focus on decentralized service delivery and community engagement. The country's health transformation plan and universal health coverage scheme (Jaminan Kesehatan Nasional) have facilitated the integration process, enabling a coordinated approach across various health-care levels. Challenges remain because of geographical diversity and varying regional priorities.

Rwanda

Rwanda has achieved remarkable progress in integrating hepatitis C services with PHC. The country's approach includes mass screening, decentralized treatment, task sharing, community engagement, high-level political commitment and sustainable financing. The integration of services has led to significant improvements, such as rapid case finding and treatment delivery and enhanced patient outcomes.

Zambia

Zambia has demonstrated robust integration efforts for HIV and noncommunicable diseases services through PHC. The country's approach involves task sharing, community engagement and collaboration with development partners. Zambia's achievements include improved access to comprehensive care and increased detection and treatment rates for coinfections. Resource constraints and coordination challenges remain areas for improvement.

Ethiopia's health-care transformation: integrating HIV, hepatitis, TB and STI services with PHC to improve outcomes

Introduction

Ethiopia has made significant strides in integrating HIV, hepatitis, TB, and STI services into PHC. The PHC programme has been implemented in Ethiopia since 2005 on the initiative of Tedros Adhanom Ghebreyesus, Minister of Health between 2005 and 2010 *(21)*. Key milestones have been achieved in advancing PHC, including the Health Extension Program introduced into the health system by training and deploying thousands of health extension workers in rural areas to provide basic health services and education at the community level: a considerable increase in health facilities (hospitals, health centres and health posts) that has improved health-care infrastructure.

Political commitment and leadership have been crucial, with supportive policies and frameworks enhancing integration efforts. Governance structures, national strategic plans, guidelines and tools facilitate coordinated action across health-care levels. Ethiopia's health system is decentralized and structured as a tiered system designed to provide services at different levels, from primary to tertiary care.

In addition, 20 years ago the Government of Ethiopia launched the Health Extension Program, a national programme for community health workers, to alleviate the severe shortage of human resources for health and improve access to PHC for the rural population. Community engagement is evident through the involvement of civil society organizations, associations of people living with HIV and community health workers in planning, implementing and monitoring integration efforts. This involvement ensures that services are tailored to the community's needs and enhances the overall effectiveness of health interventions *(21)*.

Path to integration

During strategic health planning, Ethiopia has envisioned a sustainable HIV response after 2030, with goals including universal HIV status awareness, adequate access to antiretroviral medicine and community engagement. Financial sustainability plans include integrating services into health insurance packages, local manufacturing of medications and improving financial commitments from local governments. The country also plans to continue leveraging international support while increasing domestic resource mobilization.

Ethiopia has undertaken comprehensive strategies to integrate HIV, hepatitis, TB and STI services into its PHC system. These efforts span various health-care system levels and involve multiple stakeholders, including government bodies, community organizations and international partners.

1. Governance and policy integration

- **Strategic plans.** Ethiopia has developed comprehensive national strategic plans that address the integration of HIV, hepatitis, TB and STI services. These plans emphasize the integration of these services with PHC *(22)*.

- **Coordinated governance.** The Ministry of Health coordinates these efforts through different lead executive offices. For example, the HIV, STI, TB and HIV and hepatitis programmes are coordinated under the HIV prevention and control lead executive office, and the TB programme is under the Disease Prevention and Control lead executive office of the Ministry of Health.

- **Integrated monitoring and evaluation.** The national health management information system, DHIS II, integrates the recording and reporting mechanisms for HIV, STI, hepatitis and TB programmes, ensuring streamlined monitoring and evaluation processes.

2. Strategies for integrating service delivery

- **One-stop services.** At the service delivery level, Ethiopia has implemented one-stop shopping models in which multiple services are provided at a single point of care. For instance, antiretroviral therapy clinics now offer screening, diagnosis and treatment for STI, TB and hepatitis. The HIV, TB, STI, hepatitis and sexual and reproductive health services are available at all levels of the health system. The HIV testing services are integrated with all out- and inpatient departments, TB clinics and maternal and child health services (antenatal care, family planning and clinics for children younger than five years). Antiretroviral therapy is provided in the maternal and child health platform for preventing vertical transmission. For condom promotion and distribution and post-exposure prophylaxis, they are available in all service outlets. PrEP is also available in antiretroviral therapy and antenatal care clinics.

- **Integrated package of services:** The Health Extension Program has 16 packages related to disease prevention and control (HIV, TB, malaria, STI, common injuries and emergencies) *(23)*, family health (maternal and child health, family planning, immunization, adolescent sexual and reproductive health and nutrition), hygiene and environmental sanitation (safe and adequate water supply, excretal disposal, waste management, health home environment and food hygiene and safety) and health education and communication.

- **Task sharing.** Lay health workers, including community health workers, peer champions and volunteers, are engaged in providing services. This approach enables task sharing and shifting, which is crucial in providing person-centred health services.

- **Community-based services.** People with well-controlled HIV receive follow-up services at the community level, with the services supported by community-based organizations and health extension workers. They consist of local administrative units and clubs that raise awareness about HIV, promote safe practices and support peers living with HIV. They also provide care and support to people living with HIV in their homes, helping to reduce stigma and improve health outcomes. In addition, traditional birth attendants and community health agents play a crucial role in these efforts. This decentralization of services ensures that preventive and non-clinical services are cascaded to the community while critical cases are referred to health-care facilities.

- **Health Extension Program.** The Health Extension Program, one strategy adopted by the Government of Ethiopia to achieve universal coverage of PHC among the rural population in a context of limited resources, has been pivotal in decentralizing services. About 40 000 health extension workers provide basic HIV prevention and family planning services at the household level. Recently, the health extension workers have been actively engaged in HIV service delivery through differentiated service delivery models. The contribution of the Health Extension Program to the national HIV response has mostly been limited to behavioural HIV prevention and condom promotion *(21)*.

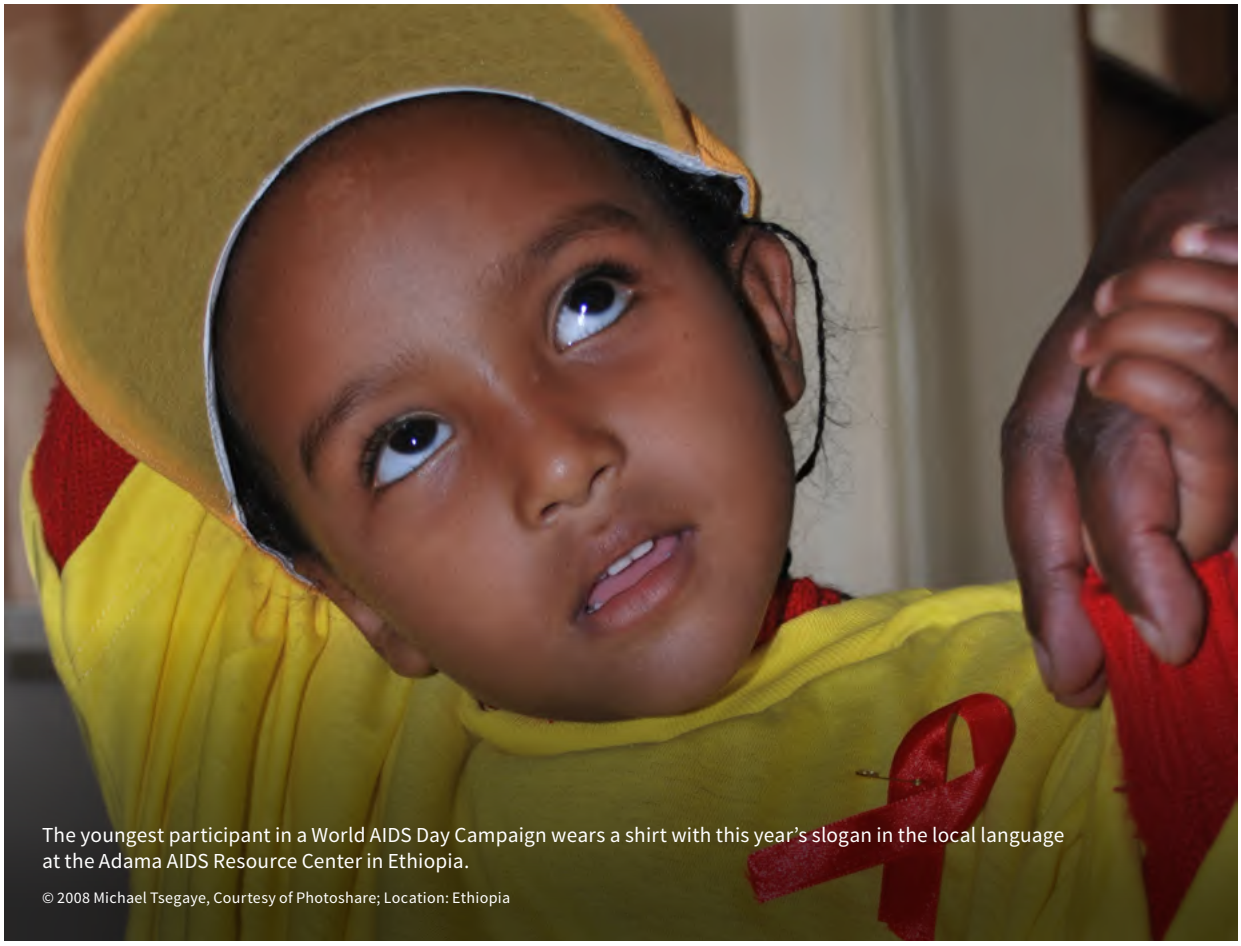
3. Resource mobilization and allocation

- **Domestic and international financing.** The integration efforts are supported through a mix of government funding and international aid. Ethiopia has developed a health-care financing strategy to increase resources, improve efficiency and ensure sustainability. International partners such as PEPFAR *(24)* and the Global Fund *(25)* have contributed significantly, although the emphasis on increasing domestic resource mobilization is growing. The exact proportions of funding from the Government of Ethiopia versus international donors can vary yearly *(26)*. Generally, international donors provide most of the funding for HIV and AIDS programmes, with estimates suggesting that donor contributions account for about 80–90% of the total HIV and AIDS funding in Ethiopia. The Government of Ethiopia is committed to addressing the epidemic but contributes a smaller share, typically 10–20%. Most domestic funding goes to workforce salaries and expanding health-care infrastructure. STI and viral hepatitis services are subsidized through domestic funding.

- **Community health insurance.** The government has implemented community-based health insurance schemes for HIV and TB to protect people from financial hardship and ensure access to integrated health-care services. The community-based health insurance scheme was introduced and is currently widely implemented in the country. The Government of Ethiopia provides subsidies to support the community-based health insurance schemes, especially for the poorest households. These subsidies help to cover the premiums for those who cannot afford to pay.

4. Community engagement and participation

- Involvement of people living with HIV and key populations. People living with HIV and key populations are actively involved in policy-designing processes, representation, advocacy and awareness, formulating strategy, implementing programmes, building capacity, delivering services and monitoring and evaluation.



The youngest participant in a World AIDS Day Campaign wears a shirt with this year's slogan in the local language at the Adama AIDS Resource Center in Ethiopia.

© 2008 Michael Tsegaye, Courtesy of Photoshare; Location: Ethiopia

- Civil society organizations. Civil society organizations and community-based organizations are integral to the integration process. Several civil society and community-based organizations play crucial roles in HIV prevention, care and treatment in Ethiopia, complementing government and international nongovernmental organization efforts. These civil society organizations enhance the national HIV response by providing direct services, advocating for policy development, supporting people living with HIV and reducing stigma and discrimination.

5. Training and capacity building

- **Continuing professional development.** Health-care providers receive regular training on integrated service delivery models. This includes in-service training on new guidelines and best practices for managing HIV, hepatitis, TB and STI services, including modules on the harmful effects of stigma and discrimination and strategies to reduce these attitudes among health-care workers.

- **Support systems.** Mentorship programmes and supportive supervision are established to ensure that health-care providers have the necessary skills and support to deliver integrated services effectively.

6. Digital and technological integration

- **Use of digital tools.** Digital technologies are used to enhance service delivery and data management. This includes electronic health records, mobile health applications and telemedicine services, which help to track patient outcomes and ensure continuity of care. The DHIS II system has integrated recording and reporting mechanisms, including HIV, STI, hepatitis and TB programmes.

- **Laboratory integration.** The integration of laboratory services includes using GeneXpert machines for both TB and HIV early infant diagnosis, demonstrating an efficient use of available resources.

Outcomes of integration efforts

Strengthening PHC through integration has improved client satisfaction, reduced morbidity and minimized health-care expenses. Preventive and curative services are effectively balanced, with community-based organizations playing a crucial role in providing preventive services and linking patients to health-care facilities for treatment. The involvement of people living with HIV and TB and key populations in the integration process has enhanced service delivery and community trust.

The impact on health outcomes includes the following:

- **Improved client satisfaction.** Patients report higher satisfaction levels because of the holistic approach to health care that addresses multiple needs in one visit. This has also led to improved adherence to treatment regimens.

- **Health system strengthening.** The integration has strengthened the health system by improving data management, resource allocation and health-care provider training. The use of digital technologies has streamlined processes and enhanced service delivery. Ethiopia has achieved significant progress against HIV, TB and malaria while building resilient and sustainable systems for health.

- **Economic benefits.** Reducing morbidity and health-care expenses has had positive economic implications. Improved health outcomes increase productivity and reduce the financial burden on families and the health-care system.

- **Capacity building.** Health-care providers have obtained new skills and knowledge, enhancing their ability to deliver high-quality care. Training and mentorship programmes have increased their motivation and job satisfaction.

- **Community engagement.** They are instrumental in facilitating community and facility linkage and in generating real-time data on the quality and availability of services through community-led monitoring. This work also helps reduce stigma and discrimination.

The path to integration has had challenges. Some lessons learned from the integration efforts include the following:

- **Resource constraints.** Despite significant progress, challenges remain in securing adequate funding, especially for establishing viral hepatitis programmes and for optimizing STI programmes.

- **Overburdened health-care providers.** Integrating multiple services has increased the workload for health-care providers, leading to potential burnout and stress. This can negatively affect the quality of care provided.

- **Infrastructure challenges.** Some health-care facilities may not have the necessary infrastructure to support integrated services, leading to inconsistencies in service delivery across regions.

- **Potential for service quality problems.** With the increased complexity of integrating multiple services, service quality risks being compromised if this is not managed properly. Ensuring continuing training and support for health-care providers is crucial to mitigate this risk.

- **Community perceptions.** Although community engagement has been positive overall, some community members may be concerned about the adequacy of integrated services, especially in areas with limited resources or high disease prevalence.

Integrating HIV, hepatitis, TB, and STI services with PHC in Ethiopia has shown promising results in improving health outcomes and system efficiency (see Fig. 10). Continued commitment, innovative strategies and sustainable funding models are essential for maintaining and advancing these gains. In Ethiopia, the collaborative efforts of the government, development partners and communities and civil society have created a robust foundation for integrated health services.

Recommendations for other countries

Lessons learned and programmatic improvements. Key lessons highlight the importance of strong political commitment, community engagement and multisectoral coordination. Recommendations for other countries include investing in resilient health and community systems and leveraging resources. Effective integration requires continual monitoring, active community engagement and adaptive strategies to address emerging challenges.

Quality of care and engagement strategies. Improving care quality involves using digital technologies and engaging private-sector providers. Community feedback mechanisms, such as client satisfaction surveys and mystery client interviews, help to address needs and concerns, enhancing the integration model. These strategies ensure that services are person-centred and responsive to community needs.

PHC levers identified for priority action in Ethiopia

PHC levers in Ethiopia	
Demonstrates evidence of political commitment and leadership	
Governance and policy frameworks are in place	
Demonstrates evidence of funding and allocation of resources	
Engagement of community and other stakeholders is referenced	
Models of care are adapted	
PHC workforce is strengthened	
Physical infrastructure is adapted or improved	
Includes a focus on strengthening access to medicines and other health products	
Evidence of engagement with private-sector providers or partners	
Purchasing and payment systems are adapted	
Includes a focus on digital technologies for health	
Systems are in place for improving the quality of care	
PHC-oriented research is included	
Monitoring and evaluation plans in place	

Context-specific innovations and lessons learned

Evidence of a mature system or an approach with lessons that can be readily applied to other country contexts



In an orphanage in Ethiopia, a Muslim woman takes care of two children whose parents died in the civil war.

© 2013 Jéssica Paula, Courtesy of Photoshare, Location: Ethiopia

Integrating HIV, viral hepatitis, TB and STI services with PHC in Indonesia

Introduction

Indonesia faces a significant HIV epidemic, with 570 000 people living with HIV as of 2023 *(27)*. Despite a decline in the number of people newly infected over the past 12 years, key populations, especially men who have sex with men, continue to experience a persistent burden of new infections. In response, Indonesia has committed to reducing the number of people newly infected with HIV infections and dying from AIDS-related causes by 2026 through a series of ambitious national health targets. This includes scaling up HIV testing, increasing antiretroviral therapy coverage and achieving viral load suppression among people living with HIV. However, integrating HIV, viral hepatitis, TB and STI services into PHC is critical for improving health outcomes and achieving universal health coverage in this vast country with more than 17 000 islands. Since the work to develop this case study ended, Indonesia has paused many programmes because of the United States Government funding freeze and cuts in 2025. For example, all community-led activities funded through the United States Agency for International Development were stopped, and prevention and linkage to treatment for about 30% of men who have sex with men in Jakarta have been affected. In addition, the expansion of PrEP programmes and a test pilot of long-acting HIV PrEP were halted.

Path to integration

Indonesia's Ministry of Health has made significant strides toward integrating HIV, hepatitis, TB and STI services into PHC. The integration began with the decentralization of health care in 2001, enabling local authorities to customize health policies and services based on district-specific needs. The implementation of the Jaminan Kesehatan Nasional scheme in 2014, Indonesia's universal health coverage initiative, further facilitated this transition by increasing access to health-care services free of user charges, including HIV care and viral hepatitis diagnosis and treatment, across the country. However, challenges such as geographical barriers, inequitable access to services and funding discrepancies at the district level remain barriers to full integration.

To overcome these barriers, the Ministry of Health invested in infrastructure, human resources and logistics to improve service delivery, especially in rural and underserved areas. In addition, the involvement of community health workers, key population networks and nongovernmental organizations, including those involved in community-led monitoring, have been crucial in reaching key populations, women of reproductive age, children and adolescents and addressing local health needs. The integration strategy also includes the development of a comprehensive health information system to streamline data reporting and reduce fragmentation across disease programmes.

Outcomes of integration efforts

The integration of HIV, hepatitis, TB and STI services into PHC has shown promising outcomes, though it is still in its early stages. Some of the positive results include:

- **Increased access to services.** According to information provided by the National HIV Programme Unit in June 2024, integration has made health care more accessible by providing HIV, hepatitis, TB and STI services at the *puskesmas* (community health centre) level, which are geographically closer to the population. This has reduced travel costs for patients and improved overall access to treatment and preventive care.

- **Improved screening and treatment.** Screening rates for HIV, syphilis and hepatitis B among pregnant women have increased. This has led to higher coverage rates in antenatal care settings and improved treatment outcomes for women living with HIV.

- **Service integration at the local level.** By 2023, about 746 *puskesmas* had begun implementing integrated PHC services, with the goal of scaling this up to 4000 by 2024. According to information provided by the Directorate of Primary Health Care in June 2024, they now offer comprehensive screening and treatment for HIV, viral hepatitis, TB and STIs, enhancing coordination and reducing fragmentation in service delivery.

- **Strengthened monitoring and evaluation systems.** The integration has allowed for the development of unified monitoring and evaluation systems, helping to reduce data fragmentation and ensuring that programme integration and health outcomes across disease programmes are tracked consistently and in an interlinked manner.

Challenges and key barriers

Despite the progress made, several challenges persist:

- **Resource allocation and funding.** Decentralized governance has led to discrepancies in resource allocation, with some districts lacking sufficient funding for HIV, hepatitis and STI services. In addition, stock-outs of essential medical supplies continue to pose a significant barrier to effective service delivery *(28).*

- **Workforce capacity.** The integration has placed increased responsibilities on health-care workers, especially at the community level. Although training programmes are underway to improve skills and reduce stigma, the shortage of trained personnel and high staff turnover remain challenges for sustaining high-quality care.

- **Stigma and discrimination.** Persistent stigma and discrimination, especially among key populations such as gay men and other men who have sex with men, sex workers and people who inject drugs, continue to hinder access to services. Addressing stigma through community education and support programmes remains a priority.



Doctors conduct circumcisions on young boys in Lamteuba village, Aceh Besar, Indonesia.
© 2014 Irwandi/AFN, Courtesy of Photoshare; Location: Aceh Besar, Indonesia

Recommendations for other countries

Based on Indonesia's experience, several recommendations can be made for countries seeking to integrate HIV, hepatitis, TB, and STI services into their PHC systems.

- **Strengthen coordination and governance.** Effective coordination at all levels – national, provincial, district and community – is crucial for ensuring that integration efforts are adequately funded and supported. Establishing clear governance structures and communication mechanisms can improve collaboration among health sectors.
- **Increase investment in the health workforce.** Investing in the recruitment, training and retention of health workers, especially in rural areas, is essential for maintaining the quality and sustainability of integrated health services.
- **Enhance logistical systems.** Addressing logistical challenges, especially in relation to the supply of essential medicines and medical supplies, is critical to avoid stock-outs and ensure the continuity of services.
- **Combat stigma and discrimination.** Targeted education and support programmes are needed to reduce stigma and discrimination within health-care settings and among communities, ensuring that key populations feel safe and supported when accessing health-care services.
- **Promote community engagement.** Involving community health workers and nongovernmental organizations in service delivery and monitoring is key to ensuring that services are client-centred and responsive to the needs of marginalized groups.
- **Promote public–private partnerships and strong engagement with the private sector.** Strengthened public-private partnerships will expand service coverage and improve the quality of care.

Conclusion and the way forward

Integrating HIV, hepatitis, TB, and STI services into Indonesia's PHC system is a critical step toward achieving universal health coverage and improving health outcomes (see Fig. 11). Although significant progress has been made, challenges related to resource allocation, workforce capacity and stigma remain. By addressing the remaining challenges, including those related to stigma and discrimination, and building on the successes of the integration process, Indonesia can serve as a model for other countries pursuing similar integration efforts, ultimately contributing to the global fight against HIV, hepatitis, TB and STIs. To ensure the sustainability of these efforts, ongoing investment in infrastructure, human resources and community engagement is essential.

PHC levers identified for priority action in Indonesia

PHC levers in Indonesia	
Demonstrates evidence of political commitment and leadership	
Governance and policy frameworks are in place	
Demonstrates evidence of funding and allocation of resources	
Engagement of community and other stakeholders is referenced	
Models of care are adapted	
PHC workforce is strengthened	
Physical infrastructure is adapted or improved	
Includes a focus on strengthening access to medicines and other health products	
Evidence of engagement with private-sector providers or partners	
Purchasing and payment systems are adapted	
Includes a focus on digital technologies for health	
Systems are in place for improving the quality of care	
PHC-oriented research is included	
Monitoring and evaluation plans in place	

Context-specific innovations and lessons learned

Evidence of a mature system or an approach with lessons that can be readily applied to other country contexts

Eliminating hepatitis C in Rwanda using a PHC approach

Introduction

Rwanda has made impressive progress in health, economic and social development, especially in health care. With a population of 13.2 million, the country has overcome numerous challenges, including recovering from the 1994 genocide and addressing health crises, including COVID-19 and Marburg outbreaks. Rwanda's health system has played a crucial role in improving life expectancy, which increased from 48.3 years in 2001 to 67.5 years in 2022 *(29)*. A key driver behind this success is the country's strong political will, effective governance and commitment to a decentralized health-care model that gives priority to person-centred care and integrated services.

Since 2010, Rwanda has focused on providing integrated treatment services for HIV, hepatitis, TB and STIs through PHC centres *(30)*. This approach has been extended to eliminating cervical cancer through human papillomavirus vaccination and cervical cancer screening and treatment, mental health and noncommunicable diseases and by contributing to reducing the hepatitis C prevalence from 4.7% in 2015 to 0.2% in 2024 *(31)*. Rwanda's commitment to integrating systems and services at all levels has created a robust health system that can serve as a model for other countries seeking to improve health-care delivery. Health-care resources have been made available through the government budget *(32)*, including resources raised through taxation. External funding remains critical, and large donors providing support to the HIV and viral hepatitis responses include the Global Fund. Donor funding is channelled into budget support rather than project support, with performance-based financing agreements in place.

Path to integration

Rwanda's journey toward integrating hepatitis C treatment into the health-care system testifies to the country's political commitment and multisectoral engagement. In 2018, the government set a national goal to eliminate hepatitis C, supported by the performance contract (Imihigo) system, which holds local governments accountable for screening targets. By 2020, Rwanda significantly reduced hepatitis C prevalence through extensive case finding, decentralizing services and introducing affordable treatment options.

A major success was the government's negotiation of reduced prices for hepatitis C commodities, lowering treatment costs to less than US$ 80 per person. This financial strategy facilitated the expansion of decentralized services, including the integration of GeneXpert platforms for viral load testing. By 2020, 13 viral load testing hubs and 82 PHC centres were operational, with all 508 health centres offering hepatitis C screening.

The decentralized model also incorporated task sharing, nurses leading treatment supported by district-level mentors. Peer educators further contributed by helping with treatment adherence, reducing stigma and ensuring patients' engagement in care. By 2024, more than 6000 health-care providers had been trained and nearly 9 million people screened for hepatitis C, resulting in treatment for more than 61 000 people.

Outcomes of integration efforts

The integration of hepatitis C services has resulted in a rapid scale-up of health-care delivery across Rwanda. Before 2017, hepatitis B and C treatment was limited to tertiary hospitals, but by 2020, all 508 health centres were equipped to offer screening, diagnosis and treatment for hepatitis B and C.

One of the key benefits of the integrated approach is improved service delivery. Patients now receive multiple services in a single visit, reducing travel and wait times and eliminating the financial burden of accessing treatment. Further, integrating services has streamlined care for hepatitis, HIV, TB and noncommunicable diseases, increasing the efficiency of health-care delivery and improving patient satisfaction based on a PHC approach.

The path to integration has had challenges:

- **Human resource constraints.** Sustaining integrated services requires a sufficient, well-trained workforce at PHC centres. Regular training, mentorship, managing staff attrition and addressing workload challenges are essential to prevent burnout to maintain high-quality care. Long-term strategies are needed to increase and retain necessary numbers of health-care workers.

- **Stigma and discrimination.** Despite efforts to reduce stigma, certain populations still face discrimination, hindering their access to care. People with health issues such as hepatitis C may feel uncomfortable receiving services in clinics that also provide HIV treatment, and this can limit the effectiveness of peer educators across disease responses.

- **Variability in service quality.** The quality of integrated services may vary across regions and facilities, leading to disparities in health outcomes. Effective quality of care monitoring and evaluation systems are needed to identify and address gaps in service delivery and their quality.

Recommendations for other countries

- **Secure political commitment and sustainable funding.** To maintain and expand integrated services, countries should secure political commitment at highest and all relevant policy and programme implementation levels and sustainable funding from both domestic and international sources. Innovative financing models, such as social health insurance and risk pooling, can help to minimize the reliance on expensive tertiary care and reduce out-of-pocket expenses for vulnerable populations.

- **Strengthen PHC-oriented systems.** Investing in PHC is essential for scaling up person-centred services. Countries should leverage disease-specific funding to build strong PHC-oriented health systems that are cost-effective and accessible to all populations. Investment approaches that leverage disease-specific funding can be used to create strong PHC systems and human resources, which in turn provide a platform for scaled up patient-centred services that are low cost and accessible.

- **Utilize task sharing and mentorship.** Task-sharing strategies, combined with continuing mentorship and training, have proven effective in extending health-care reach. A capacity-building system that incorporates both in-person and e-learning approaches will help to ensure that health-care workers are well-equipped to deliver integrated services.

- **Foster multisectoral engagement.** Engaging all community-level actors is crucial for the success of integrated services. Countries should foster multisectoral partnerships and accountability mechanisms to support health goals and ensure the active participation of all stakeholders.

- **Institutionalize joint planning and coordination.** Effective coordination platforms are essential for the smooth implementation of integrated services. Institutionalizing joint planning and using data-driven approaches can reduce duplication, improve alignment with national goals and harmonize guidelines, procurement and reporting systems across programmes.

Conclusion

Rwanda's integration of hepatitis C services into PHC has been highly successful, building on the infrastructure and lessons learned from its HIV response (Fig. 12). Political commitment, strong leadership, effective governance, innovative financing, and community engagement have been key to this achievement. Moving forward, countries seeking to replicate Rwanda's success must focus on elevation of political commitment and leadership, addressing resource constraints, improving service delivery models and reducing stigma to ensure sustainable, long-term health outcomes. Rwanda's experience serves as a valuable model for countries striving to improve health-care delivery through integrated services.

PHC levers identified for priority action in Rwanda

PHC levers in Rwanda	
Demonstrates evidence of political commitment and leadership	
Governance and policy frameworks are in place	
Demonstrates evidence of funding and allocation of resources	
Engagement of community and other stakeholders is referenced	
Models of care are adapted	
PHC workforce is strengthened	
Physical infrastructure is adapted or improved	
Includes a focus on strengthening access to medicines and other health products	
Evidence of engagement with private-sector providers or partners	
Purchasing and payment systems are adapted	
Includes a focus on digital technologies for health	
Systems are in place for improving the quality of care	
PHC-oriented research is included	
Monitoring and evaluation plans in place	

Context-specific innovations and lessons learned

Evidence of a mature system or an approach with lessons that can be readily applied to other country contexts

Strengthening PHC in Zambia: integrating HIV, viral hepatitis, TB and STI services and noncommunicable diseases with PHC for holistic person-centred care

Introduction

Zambia has demonstrated strong political commitment and leadership in integrating HIV, viral hepatitis, TB and STI services into PHC. The National Health Strategic Plan and disease-specific frameworks such as the National HIV/AIDS Strategic Framework emphasize comprehensive and integrated service delivery. These frameworks align with broader Sustainable Development Goals aiming to achieve universal health coverage *(33).*

Zambia's health-care system is organized into a tiered structure designed to provide comprehensive care. Health posts and community health workers deliver essential primary care and preventive services at the community level. Health centres and district hospitals comprise the PHC facilities, offering more advanced services and acting as referral points for community health services. Provincial and tertiary hospitals provide specialized care and serve as referral centres for district hospitals equipped with advanced medical technologies and specialist staff.

Community-based organizations, community-led organizations and nongovernmental organizations significantly reach underserved populations through mobile clinics and outreach programs. These efforts promote equity in access to health-care services and ensure that community needs are addressed. Community engagement is facilitated through stakeholder meetings, community advisory boards and participatory planning, ensuring that the voices of key populations are included in decision-making processes.

Path to integration

Zambia has envisioned a sustainable HIV response after 2030 through advances in medical technology, integration of services and community-led initiatives. Financial sustainability plans included increased government funding, health insurance schemes and innovative financing mechanisms such as social impact bonds. Ensuring sustainable financing models and integrating services within universal health coverage frameworks are critical.

The National Health Insurance Scheme was launched in 2019 to provide universal health coverage to all Zambians. Managed by the National Health Insurance Management Authority, the Scheme offers a range of health-care benefits, including access to essential medicines and outpatient and inpatient services. The National Health Insurance Scheme covers many services, including treatment for HIV-related health problems, although specific details may vary based on the policy tier *(34).* Antiretroviral therapy is often available free of charge through the Ministry of Health and international partners, such as PEPFAR and the Global Fund. This support reduces the financial burden on the National Health Insurance Scheme for HIV treatment, since antiretroviral therapy is mostly provided through government and donor-funded programmes.

An essential step to integration was creating differentiated models of care and workforce development. Integration efforts included establishing one-stop clinics and using mobile units for outreach in remote areas. Health-care facilities are structured to provide various services under one roof, reducing the need for patients to visit multiple clinics. Task-sharing strategies involve community health workers and volunteers to deliver comprehensive care *(35)*. Training programmes are conducted to equip health-care workers with the skills necessary for managing multiple diseases and promoting high-quality, people-centred primary care.

Zambia used the task shifting already occurring in HIV services to address growing noncommunicable diseases among people living with HIV. The resources available for HIV service delivery settings were leveraged to enable the integration of noncommunicable diseases, including electronic medical record systems, which were adapted to include noncommunicable disease modules and strengthened supply chains to support consistent supply of medicines and other commodities for noncommunicable diseases. These lessons can be applied to populations other than those affected by HIV. Task shifting to community health workers was also used to manage both HIV and noncommunicable diseases. They play a key role in education, follow-up and adherence counselling after their training, especially in rural and underserved areas.

Engaging key populations and communities in planning and executing integrated services was essential. Community advisory boards and participatory planning sessions have been instrumental in ensuring that the needs and perspectives of people living with HIV and key populations are considered in policy and programme decisions, which has led to improved service uptake and reduced stigma.

Collaboration with development partners, including WHO, UNAIDS and various nongovernmental organizations, was necessary to align their support with national policies and community needs. These partners provided technical assistance, funding and advocacy to enhance the integration process and ensure sustainability. They support integrated health initiatives by offering comprehensive service packages, supporting policy development, community engagement and facilitating multisectoral coordination.

Outcomes of integration efforts

A focus on effectiveness and balancing services by focusing on integration has improved service delivery, balancing preventive and treatment services within health-care facilities and community-based organizations. People living with HIV, key populations, and vulnerable populations including adolescents are actively engaged through community empowerment and sensitizations and peer education programmes. The perceptions of these populations towards integration efforts are generally positive, citing reduced stigma and improved access to comprehensive care and other services.

Many health-care facilities in Zambia, especially at the primary care level, are starting to provide integrated HIV and noncommunicable disease services. Routine HIV care now often includes screening for hypertension, diabetes and other noncommunicable diseases *(36)*. This reduces the need for attending multiple appointments, improving adherence to treatment and reducing the burden on health-care workers.

The impact on health outcomes includes the following:

- **Enhanced access to services.** By integrating services, Zambia has made it easier for individuals to access comprehensive health care. This includes integrating HIV, TB, noncommunicable diseases, mental health, family planning, and sexual and reproductive health services, which ensures that patients receive holistic care *(37)*. The Cervical Cancer Prevention Program in Zambia is a notable example of integrated care. Initially targeting women living with HIV, the Program expanded to serve all women across 33 health-care facilities nationwide *(38)*. Regarding the prevention of vertical transmission of HIV and hepatitis B, integrating vaccination services for the hepatitis B birth dose, which achieved 94% coverage in 2019, was crucial. The number of people who avoided becoming newly infected with HIV by preventing vertical transmission was 9500 in 2023 *(39)*.

- **Increased detection and treatment.** Integration has led to more comprehensive screening and timely treatment of coinfections such as TB among people living with HIV, improving overall health outcomes *(40)*.

- **Reduced transmission.** Enhanced adherence to treatment regimens and comprehensive care have contributed to lowering the transmission rates of communicable diseases.

- **Improved service delivery.** Patients benefit from accessing multiple services in one visit, reducing travel and wait times. This streamlined approach enhances the efficiency of health-care delivery and patient satisfaction.

- **Holistic care.** A holistic approach addresses multiple health issues simultaneously, improving the overall quality of care.

- **Better health outcomes.** The integration approach has led to better health outcomes, including higher rates of disease detection, improved treatment adherence and reduced transmission rates of HIV, hepatitis and STIs. Integrating HIV and noncommunicable disease services has led to better health outcomes for people living with HIV and noncommunicable diseases. This approach addresses comorbidities such as cardiovascular disease, diabetes and cancer, which are common among people living with HIV.

- **Increased efficiency.** Integration has streamlined health-care delivery, reduced duplication of efforts and improved the use of resources. This has led to more efficient service delivery and improved health outcomes.

The path to integration has had challenges. Some key lessons learned include the following:

- **Resource constraints.** Sustaining integrated services requires significant financial investment, which can be challenging. Limited financial and human resources can hinder the scaling up of integrated services. Rural populations may face access challenges to both National Health Insurance Scheme and HIV-specific health-care services because of limited health-care facilities and workforce shortages. Advocacy for increased domestic funding and exploring innovative financing mechanisms are necessary.

- **Health workforce issues.** Health-care providers may experience increased workloads, leading to potential burnout and affecting the quality of care.

- **Infrastructure and logistics.** Some health-care facilities struggle to accommodate the increased client load, resulting in overcrowding and longer wait times. Ensuring a consistent supply of medications and diagnostic tools for multiple infections can also be challenging.

- **Coordination and implementation challenges.** Effective integration requires strong coordination among health programmes and departments. Poor coordination can lead to fragmented services and inefficiency. Continuing training and capacity-building are necessary to update health-care providers on the latest treatment protocols for multiple infections. Inadequate training can affect the quality of integrated services.

- **Variability in service quality.** The quality of integrated services can vary between regions and facilities, leading to disparities in health outcomes. Robust systems for monitoring and evaluating the quality and effectiveness of integrated services are essential. Weaknesses in these systems can create challenges in identifying and addressing gaps in service delivery.

- **Stigma and discrimination.** Despite efforts to reduce stigma, some people living with HIV and key populations still experience discrimination from health-care providers. This can deter individuals from seeking care and undermine the benefits of integrated services.

Integrating HIV, hepatitis, TB, and STI services and noncommunicable disease into Zambia's PHC system has shown significant promise in improving health outcomes and service delivery. Continued political commitment, innovative financing and strong community engagement are essential for sustaining these efforts. Future strategies should focus on addressing resource constraints, improving coordination and reducing stigma to ensure the long-term success of integrated health services.

Recommendations for other countries

- Enhance integration efforts, especially in funding and service delivery through PHC. Expanding national health insurance scheme coverage and integrating HIV and other communicable and noncommunicable disease services into general health care continue to be crucial steps toward universal health coverage.
- Strengthen health systems and continually train health-care workers to deliver integrated services effectively.
- Use electronic health records and mobile health solutions to improve service delivery and client monitoring. These technologies help to track patient history across diseases, enhancing the continuity of care.
- Foster strong community engagement to ensure that the needs and perspectives of people living with HIV and key populations are considered.
- Improve the coordination and harmonization of guidelines, protocols and reporting systems across programmes.
- Address stigma and discrimination to improve service access and uptake and reduce health disparities.
- Secure sustainable funding from both domestic and international sources to maintain and expand integrated services.
- Engage the private sector through public–private partnerships to support delivering integrated health services. This can help to optimize resource use and expand service coverage and improve the quality of services.

Zambia has demonstrated significant progress, and continued political commitment, innovative financing and strong community engagement are essential for sustaining these efforts (Fig. 13). Future strategies should focus on addressing resource constraints, improving coordination and reducing stigma to ensure the long-term success of integrated health services.

PHC levers identified for priority action in Zambia

PHC levers in Zambia	
Demonstrates evidence of political commitment and leadership	
Governance and policy frameworks are in place	
Demonstrates evidence of funding and allocation of resources	
Engagement of community and other stakeholders is referenced	
Models of care are adapted	
PHC workforce is strengthened	
Physical infrastructure is adapted or improved	
Includes a focus on strengthening access to medicines and other health products	
Evidence of engagement with private-sector providers or partners	
Purchasing and payment systems are adapted	
Includes a focus on digital technologies for health	
Systems are in place for improving the quality of care	
PHC-oriented research is included	
Monitoring and evaluation plans in place	

Context-specific innovations and lessons learned

Evidence of a mature system or an approach with lessons that can be readily applied to other country contexts

Conclusions

The country examples described in Part One and the country case studies in Part Two offer important insights into the importance of taking deliberate strategic and simultaneous action across multiple strategic and operational levers of the Operational Framework for Primary Health Care. Annex 1 provides a narrative summary of the strengths and best practices across the four case studies.

All the case studies in Part Two combined a focus on all four of the PHC operational strategic levers, with a strong focus on models of care, PHC workforce strengthening and included monitoring and evaluation (Fig. 14). These areas offered lessons that can readily be adapted to other country contexts. They also included a focus on at least three additional operational levers for additional focus – specific to local contexts.

PHC levers identified for action in Ethiopia, Indonesia, Rwanda and Zambia

	PHC levers	Ethiopia	Indonesia	Rwanda	Zambia
Strategic levers	Political commitment and leadership	■	■	■	■
	Governance and policy frameworks	■	■	■	■
	Funding and allocation of resources	■	■	■	■
	Engagement of community and other stakeholders	■	■	■	■
Operational levers	Models of care	■	■	■	■
	PHC workforce	■	■	■	■
	Physical infrastructure		■	■	■
	Medicines and other health products			■	
	Engagement with private-sector providers or partners		■		■
	Purchasing and payment systems	■	■		■
	Digital technologies for health	■	■		
	Systems for improving the quality of care		■	■	■
	PHC-oriented research				
	Monitoring and evaluation	■	■	■	■

■ Context-specific innovations and lessons learned

■ Evidence of a mature system or an approach with lessons that can be readily applied to other country contexts

Acting on only one or two areas will limit the impact and reach for these complex ecosystems. Success in countries will have the greatest sustainable impact when four or more areas are given priority at the same time – including a deliberate mix of both strategic and operational levers.

The following summary recommendations are proposed from the lessons described in the country examples and country case studies.

Recommendations

- **Enhance coordination and governance, political commitment and leadership.** Establish strong coordination and governance structures at all levels to ensure alignment with national goals. Support and encourage political leaders to engage in discussions about the evolution and financing of health programmes.
- **Secure sustainable funding.** Obtain sustainable funding from both domestic and international sources to maintain and expand integrated services. Employ innovative financing mechanisms such as community-based health insurance schemes and negotiate reduced costs for treatments.
- **Focus on health systems.** Invest consistently in PHC infrastructure and human resources to support comprehensive care. Use established health platforms (such as HIV treatment) to implement new disease interventions.
- **Use task sharing.** Implement task-sharing strategies and ensure continuing mentorship and training of health and care workers. Train health and care workers and community health workers to deliver integrated services, optimizing resource use and extending service reach.
- **Foster community engagement.** Engage affected communities in planning and providing integrated services to improve service uptake and reduce stigma as well as community-led monitoring to improve service quality.
- **Reduce stigma and discrimination.** Implement targeted education and training of health care workers and support programmes for communities to address stigma and discrimination within health and community settings.
- **Leverage digital technologies.** Use electronic health records, mobile and other digital health solutions to improve service delivery and uptake and service quality monitoring.
- **Engage the private sector.** Promote public–private partnerships to expand service coverage and improve the quality of care and other services.

Integrating HIV, hepatitis, TB, and STI services with PHC-oriented health systems in Angola, Botswana, Brazil, Ethiopia, Kenya, Indonesia, Pakistan, Rwanda, Viet Nam and Zambia has shown promise in improving health outcomes and service delivery in very different contexts.

References[2]

1. Global health sector strategies on, respectively, HIV, viral hepatitis and sexually transmitted infections for the period 2022–2030. Geneva: World Health Organization; 2022 (https://iris.who.int/handle/10665/360348). Licence: CC BY-NC-SA 3.0 IGO.

2. The impact of suspensions and reductions in official development assistance on HIV, hepatitis and STIs. Geneva: World Health Organization; 2025 (https://www.who.int/publications/m/item/the-impact-of-suspensions-and-reductions-in-official-development-assistance-on-health-systems). Licence: CC BY-NC-SA 3.0 IGO.

3. Implementing the global health sector strategies on HIV, viral hepatitis and sexually transmitted infections, 2022–2030: report on progress and gaps 2024, second edition. Geneva: World Health Organization; 2024 (https://iris.who.int/handle/10665/378246). Licence: CC BY-NC-SA 3.0 IGO.

4. WHO, UNICEF. Operational framework for primary health care: transforming vision into action. Geneva: World Health Organization; 2020 (https://apps.who.int/iris/handle/10665/337641). Licence: CC BY-NC-SA 3.0 IGO.

5. Integrating HIV programs into primary health care systems Practical considerations for country decision-makers. Sandton, South Africa: Genesis Analytics; 2024 (https://genesis.imgix.net/uploads/files/5.-GENESIS_Considerations-for-Integration_30-Sept-2024-1.pdf).

6. Evaluation of the contribution of the UNAIDS Joint Programme to strengthening HIV and primary health care outcomes. Geneva: UNAIDS, 2024 (https://www.unaids.org/en/resources/documents/2024/evaluation-hiv-primary-health-care).

7. Evaluation of the WHO Special Programme on Primary Health Care: report. Geneva: World Health Organization; 2024 (https://www.who.int/publications/m/item/preliminary-evaluation-of-the-who-special-programme-on-primary-health-care--evaluation-report).

8. Angola. In: Evaluation of the contribution of the UNAIDS Joint Programme to strengthening HIV and primary health care outcomes. Geneva: UNAIDS, 2024 Geneva: UNAIDS; 2024 (https://www.unaids.org/sites/default/files/media/documents/angola-evaluation-hiv-primary-health-care_en.pdf).

9. Botswana. In: Evaluation of the contribution of the UNAIDS Joint Programme to strengthening HIV and primary health care outcomes. Geneva: UNAIDS, 2024 (https://www.unaids.org/sites/default/files/media/documents/botswana-evaluation-hiv-primary-health-care_en.pdf).

10. Pakistan. In: Evaluation of the contribution of the UNAIDS Joint Programme to strengthening HIV and primary health care outcomes. Geneva: UNAIDS, 2024 (https://www.unaids.org/sites/default/files/media/documents/pakistan-evaluation-hiv-primary-health-care_en.pdf).

11. Sharing knowledge for stronger primary health care: Brazil. Geneva: Universal Health Coverage Partnership; 2020 (https://extranet.who.int/uhcpartnership/story/sharing-knowledge-stronger-primary-health-care).

12. Pinto P, Silva A, Lua I, Jesus G, Magno L, Santos C et al. The effect of primary health care on AIDS: a cohort study of 3.4 million individuals in Brazil. 12th IAS conference on HIV Science, 23–26 July 2023, Brisbane, Australia (https://programme.ias2023.org/Abstract/Abstract/?abstractid=3078).

13. Primary health care. London: PATH; 2025 (https://www.path.org/who-we-are/programs/primary-health-care).

14. Kenya HIV estimates portal [online database]. Nairobi: National Syndemic Diseases Control Council; 2025 (https://analytics.nsdcc.go.ke/estimates/#!/kenya/county).

15. PATH, 2024. Advancing integrated person-centered HIV and primary health care in Kenya (https://www.path.org/our-impact/articles/hiv-ncd-integration).

16. Green KE, Nguyen LH, Phan HTT, Vu BN, Tran MH, Ngo HV et al. Prepped for PrEP? Acceptability, continuation and adherence among men who have sex with men and transgender women enrolled as part of Vietnam's first pre-exposure prophylaxis program. Sex Health. 2021;18(1):104–15 (https://doi.org/10.1071/SH20167).

17. Vu BN, Green K, Phan H, Tran L, Phan P, Tran T et al. Engaging the private sector as part of HIV pre-exposure prophylaxis service delivery in Vietnam: a comparative analysis of uptake, persistence and HIV seroconversion from 2018 to 2023. Front Reprod Health. 2025;6:1439461 (https://doi.org/10.3389/frph.2024.1439461).

18. Vu BN, Tuan KD, Tran AK, Tran LK, Green K, Nguyen KT et al. Community-based and HIV integrated testing for hepatitis B and C among key populations in Vietnam. Clin Liver Dis (Hoboken). 2022;19(4):131–7 (https://doi.org/10.1002/cld.1221).

19. Doan AH, Vu CMH, Nguyen TT, Green KE, Phan HTT, Janamnuaysook R et al. Caring for the whole person: transgender-competent HIV pre-exposure prophylaxis as part of integrated primary healthcare services in Vietnam. J Int AIDS Soc. 2022;25 Suppl. 5(Suppl. 5):e25996 (https://doi.org/10.1002/jia2.25996).

20. USAID/PATH Support for Technical Excellence and Private-Sector Sustainability in Vietnam: Final Performance Report December 10, 2021–March 26, 2025.

21. Realizing universal health coverage through primary health care: a roadmap for optimizing the Ethiopian Health Extension Program 2020–2035. Addis Ababa: Federal Ministry of Health, Ethiopia; 2023.

22. National Strategic Plan for HIV/AIDS 2023–2027. Addis Ababa: Federal Ministry of Health, Ethiopia; 2023 (https://hivpreventioncoalition.unaids.org/en/resources/ethiopia-hivaids-national-strategic-plan-202324-202627).

23. Ethiopia malaria funding request. Addis Ababa: Federal Ministry of Health, Ethiopia; 2023.

2 All references were accessed on 2 June 2025.

24. Ethiopia country operational plan (COP) 2022 strategic direction summary, April 15, 2022. Washington (DC): PEPFAR; 2022 (Ethiopia-COP22-SDS.pdf).

25. Ethiopia and Global Fund launch new grants to accelerate progress against HIV, TB and malaria; strengthen health systems. Geneva: Global Fund to Fight AIDS, Tuberculosis and Malaria; 2024 (https://www.theglobalfund.org/en/updates/2024/2024-06-18-ethiopia-global-fund-new-grants-accelerate-progress-hiv-tb-malaria-strengthen-health-systems).

26. Federal Ministry of Health, Ethiopia RSSH funding request. Geneva: Global Fund to Fight AIDS, Tuberculosis and Malaria; 2023 (https://data.theglobalfund.org/location/ETH/access-to-funding).

27. Status of HIV programmes in Indonesia. Geneva: UNAIDS, 2025 (https://www.unaids.org/en/resources/presscentre/featurestories/2025/february/20250224_indonesia_fs#:~:text=There%20are%20an%20estimated%20570%2C000%20people%20living%20with,are%20accessing%20treatment%20and%2014%25%20are%20virally%20suppressed).

28. Rakmawati T, Hinchcliff R, Pardosi JF. District-level impacts of health system decentralization in Indonesia: systematic review. Int J Health Plann Manage. 2019;34(2):e1026–53 (https://doi.org/10.1002/hpm.2768).

29. WHO Global Health estimates (https://www.who.int/data/gho/data/indicators/indicator-details/GHO/life-expectancy-at-birth-(years)).

30. Nsanzimana S, Prabhu K, McDermott H, Karita E, Forrest JI, Drobac P et al. Improving health outcomes through concurrent HIV program scale-up and health system development in Rwanda: 20 years of experience. BMC Med. 2015;13:216 (https://doi.org/10.1186/s12916-015-0443-z).

31. Hepatitis national data. Ministry of health Rwanda.

32. UNICEF health budget brief Rwanda, 2023–2024. New York: United Nations Children's Fund; 2023 (https://www.unicef.org/rwanda/media/5736/file/UNICEF_202324%20Health%20Budget%20Brief_layout_corrections_25-3-24.pdf.pdf).

33. National Health Strategic Plan 2017–2021. Lusaka: Ministry of Health, Zambia; 2017 (https://www.tbdiah.org/resources/publications/zambia-national-health-strategic-plan-2017-2021).

34. Mazimba Z, Muwowo EC, Chilufya KM, Malumani M. The National Health Insurance Scheme (NHIS) and the attainment of universal health coverage in Zambia. Rwanda Public Health Bull. 2023;4(1) (https://www.ajol.info/index.php/rphb/article/view/273914).

35. Okoroafor SC, Christmals CD. Task shifting and task sharing implementation in Africa: a scoping review on rationale and scope. Healthcare (Basel). 2023;11(8):1200 (https://doi.org/10.3390/healthcare11081200).

36. Integrating HIV and hypertension care for enhanced health outcomes. Geneva: World Health Organization; 2024 (https://www.who.int/publications/m/item/integrating-hiv-and-hypertension-care-for-enhanced-health-outcome).

37. Accelerating the integration of HIV and RMNCH services. Experiences from USAID Zambia integrated health. PEPFAR and USAID technical brief. Zambia. Washington (DC): PEPFAR; 2023 (https://www.pactworld.org/sites/default/files/2024-07/AIDS%202024%20GH%20factsheet_ZIH.pdf).

38. Long, full, healthy lives: delivering on the commitment to integrated noncommunicable disease care for people living with HIV by 2025. Geneva: NCD Alliance; 2021 (https://ncdalliance.org/resources/long-full-healthy-lives-delivering-on-the-commitment-to-integrated-ncd-care-for-people-living-with-hiv-by-2025).

39. Global HIV and AIDS statistics. Geneva: UNAIDS; 2024 (https://www.unaids.org/en/resources/fact-sheet).

40. Impact of HIV and TB integrated care: Zambia. Geneva: UNAIDS; 2021 (https://open.unaids.org/countries/zambia).

Annex 1

Summary of best practices and outcomes from the four country case studies

Country	Strengths	Best practices and outcomes
Ethiopia	Integrated HIV, TB viral hepatitis and STIs system and services Health Extension Program for rural populations The Health Extension Program and community-based health insurance schemes Strong political commitment and coordinated governance have facilitated effective integration	Ethiopia's Health Extension Program trains workers to provide basic health services at the community level, improving access in rural areas
Indonesia	Integrated HIV, TB, STIs and viral hepatitis systems and services Decentralized health system and extensive PHC facilities Extensive health transformation plan and universal health coverage scheme (Jaminan Kesehatan Nasional) Geographical diversity and varying regional priorities present unique challenges	Indonesia's decentralization policy enables customized health planning at the district level, facilitating efficient service delivery across its vast geography
Ethiopia	Integration of hepatitis C treatment into the health-care system Leveraging existing infrastructure for other diseases such as HIV High-level political commitment and mass screening efforts for hepatitis C are exemplary The success in rapid case finding and treatment delivery	Use of its established HIV treatment platforms to implement hepatitis C screening and treatment, achieving rapid case finding and treatment initiation
Zambia	Integrating HIV services with noncommunicable diseases Integrating HIV services with noncommunicable diseases is a unique approach Collaboration with development partners and robust community engagement are strengths	Zambia's integration of noncommunicable disease services with routine HIV care addresses comorbidities and improves health outcomes for people living with HIV

World Health Organization
20 Avenue Appia
1211 Geneva 27
Switzerland
who.int